

Forecasting US Housing Construction

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Introduction

Housing affordability is a widespread problem in the United States. Many Americans find homeownership prohibitively expensive. A key part of the American dream is out of reach for a sizable proportion of the population. Driving these high prices is a failure to build enough homes to meet demand. Forecasting housing construction can thus provide insight into the housing affordability crisis. This report forecasts monthly housing starts with three models: ETS, ARIMA, Regression with ARIMA errors. It first decomposes housing starts time series. Then it splits a time series of housing starts from the Census Bureau into a training set and a test set. A year was chosen as a cutoff. Observations before the cutoff year were assigned to the training set with the rest assigned to the test set. The training set was used to construct the models. Residuals and the Ljung-Box test results for each model are reported. Model accuracy is then predicted by comparing model predictions with the actual data in the test set. To account for the impact different splits have on the predicted model accuracy, models were constructed for a cutoff of 2006, 2012, 2018. The cutoffs created approximate train/test splits of 70/30, 80/20, 90/10. Lastly, the models are compared and then used to forecast housing starts two years into the future.

Datasets

Monthly housing starts are taken from the Census Bureau Survey of Construction (SOC). The SOC provides national and regional statistics on new residential construction. Housing starts are defined as the number of single family and multifamily units that started construction. The SOC defines starting construction as when excavation begins for the foundation of the building. The dataset records observations from January 1959 to March 2025. The US Bureau of Labor Statistics time series of seasonally adjusted monthly unemployment rate and producer price index for construction materials, the Federal Reserve time series on the monthly Federal Funds Effective Rate (Average of daily rate), and the S&P Case-Shiller index for national home prices were used as predictors for the regression model. The predictors for the regression model were retrieved from Federal Reserve Economic Data (FRED).

Methodology

Train/Test Sets

The dataset on housing starts was divided into a training and test set. Three different years were used as cutoffs: 2006, 2012, 2018. These cutoffs resulted in an approximate 70/30, 80/20, 90/10 train/test split. This was done to account for the decline in housing starts during the recession. Models trained on data before the recession will likely fail to forecast the decline. While models trained on data after the crash but before the increase will likely fail to forecast the recovery in housing starts. Model accuracy will vary depending on the split. To record this impact, three splits corresponding to these scenarios were created. 2006 is before the recession, 2012 before the recovery, 2018 after the recovery started.

ETS/ARIMA Models

The parameters for the ETS model were calculated using the ETS function. Parameters for the ARIMA model were chosen with the ARIMA function with stepwise and approximation set to false in order to increase the number of models compared. Residual diagnostics for each model were created. A Ljung-Box test was

also reported. No transformation was performed on the data as variation in housing starts did not vary as time increased. The model predictions were then compared with the test set. The RMSE, MAE, MAPE, MASE, RMSSE are calculated. Lastly, one year forecasts were created.

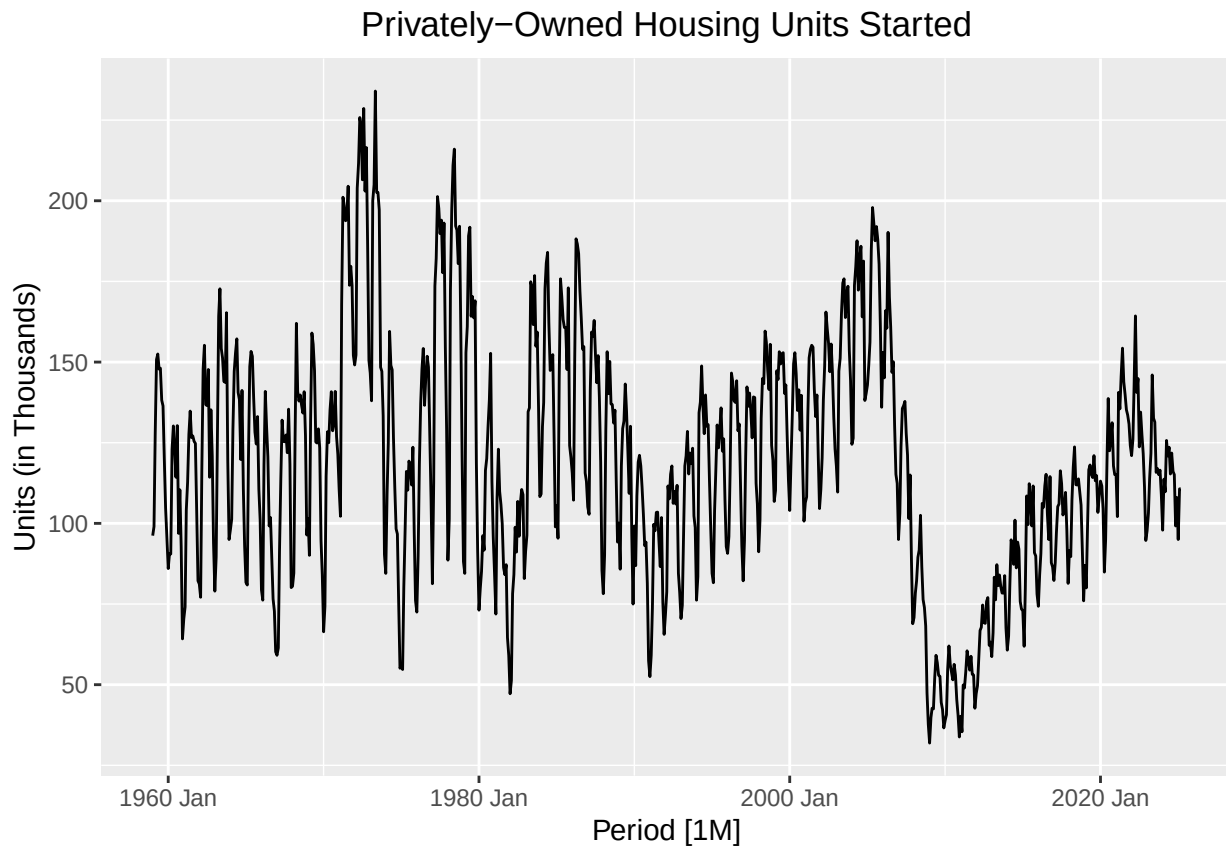
Regression with ARIMA errors

ARIMA errors are predicted with the ARIMA function. The following predictors were used: unemployment rate, federal funds effective rate, producer price index for construction, and the Case-Shiller index for national home prices. The unemployment rate was chosen to represent the state of the economy. Terrible economic conditions likely led to less homes being constructed and boom times led to more housing construction. Unemployment was chosen instead of GDP as it is recorded monthly instead of quarterly. This matches the housing start data. Federal funds rate corresponds with interest rates. Higher interest rates curb housing construction by increasing the costs of loans. PPI represents the cost to build a home. Case-Shiller represents change in total single family home prices. Higher prices incentivized home builders to build more. The predictors and housing starts were added to one csv file. The observation dates started in January 1959 and continued on to March 2025. Unfortunately, the Case-Shiller only includes data from 1987 to January 2025. Including the index decreases the size of the training set and possibly the accuracy of the model. For this reason, a regression was conducted without the Case-Shiller index and one with the index to be evaluated on the test set separately. In addition, a regression with the predictors lag by a month and a regression with the lagged predictors and the Case-Shiller index were constructed. Lagged predictors may be better at predictions since they are actually observed by home builders. The regression AICc are then compared. The best regression without the Case-Shiller index and the best regression with the Case-Shiller residuals were reported and then were evaluated with the test set. The RMSE, MAE, MAPE, MASE, RMSSE are calculated for both models.

To calculate the forecast of the two regression models, three scenarios for the predictors were chosen. It is hard to forecast the predictors. For this reason, extreme scenarios were chosen. This provides a wide range of potential outcomes. The first scenario, the predictors follow a drift model. This represented little change to the predictors. Second, unemployment rate, federal funds rate, and the producer price index decrease at a constant rate until they reach their recent minimum value. For lagged predictors, this meant the current month value or next month lagged value was the minimum value. The Case-Shiller index increased at a constant rate that matched its recent highest yearly increase. This represented a scenario where the predictors changed in a way associated with more housing starts. Third, the federal funds rate, the producer price index, and the Case-Shiller index increased at a constant rate until they reached their recent maximum value. Unemployment rate increased until its second largest recent value to avoid using the COVID outlier. This represented a scenario where the predictors changed in a way associated with less housing starts. Unfortunately, some NaN values were created, so the regression used to forecast future values lacked certain parameters.

Time Series Decomposition

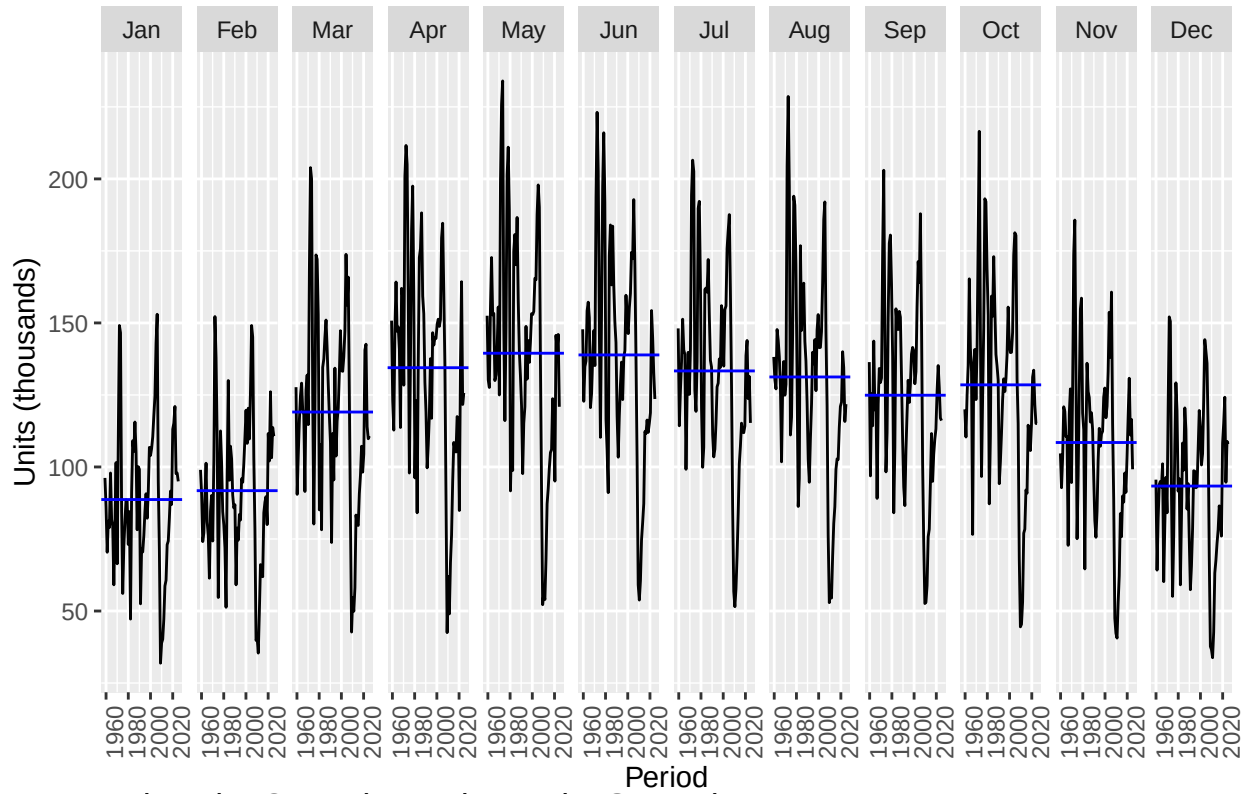
Trend/Cycle



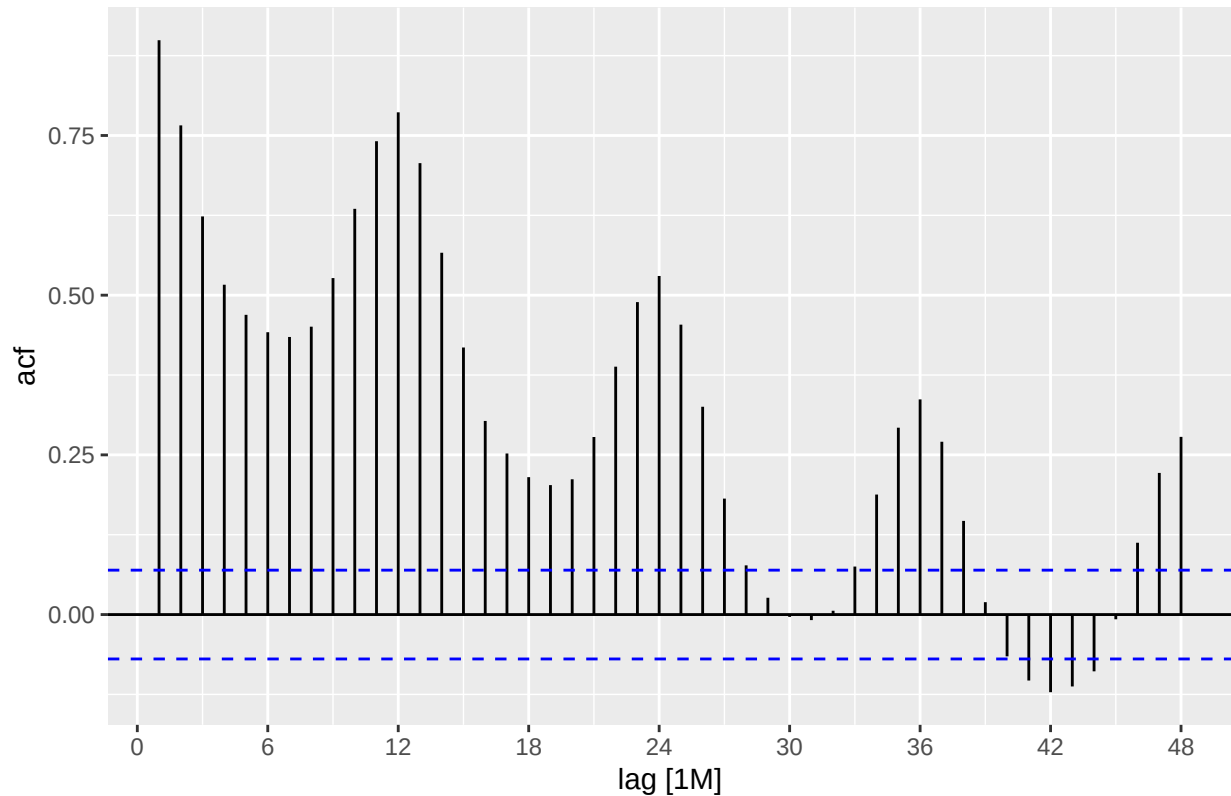
The time series has no overall trend but it does show cyclical behavior. Most obviously, there is a peak around 2006 and 2007 before a drastic decline around 2008. This of course represents the housing bubble and the 2008 recession.

Seasonality

Seasonal plot: Privately-Owned Housing Units Started



Privately-Owned Housing Units Started

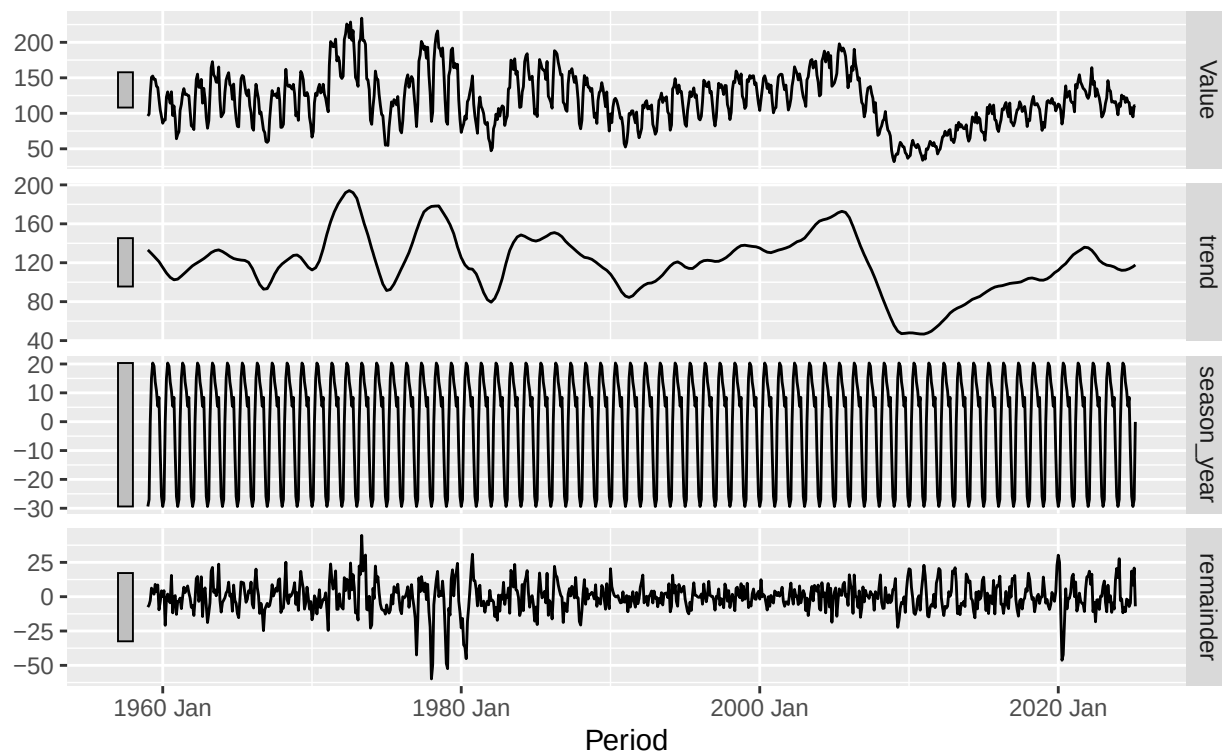


There is also clear seasonality. The seasonal subplots show more housing starts in the summer months than winter. Further, the ACF peaks every 12 months, a clear sign of seasonality.

STL Decomposition

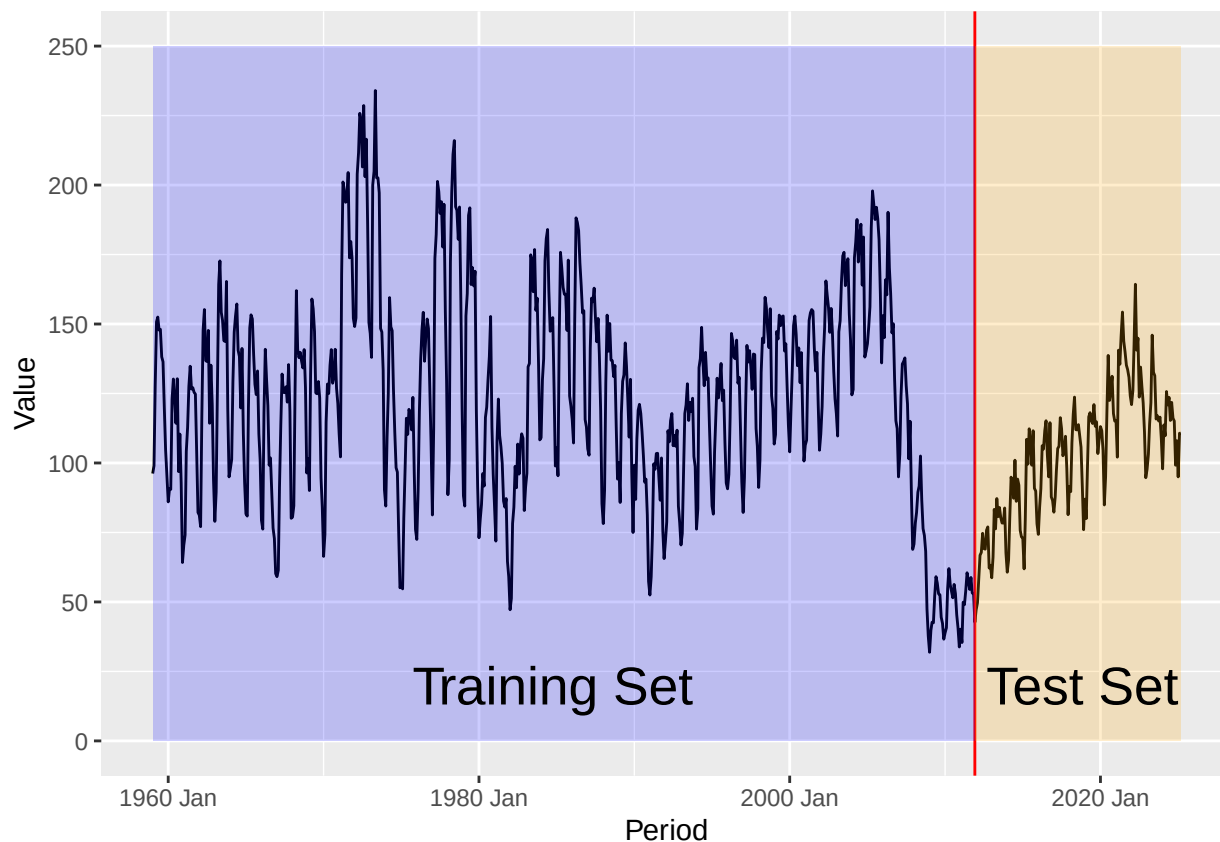
STL decomposition

Value = trend + season_year + remainder



Models

80/20 Training/Test Set

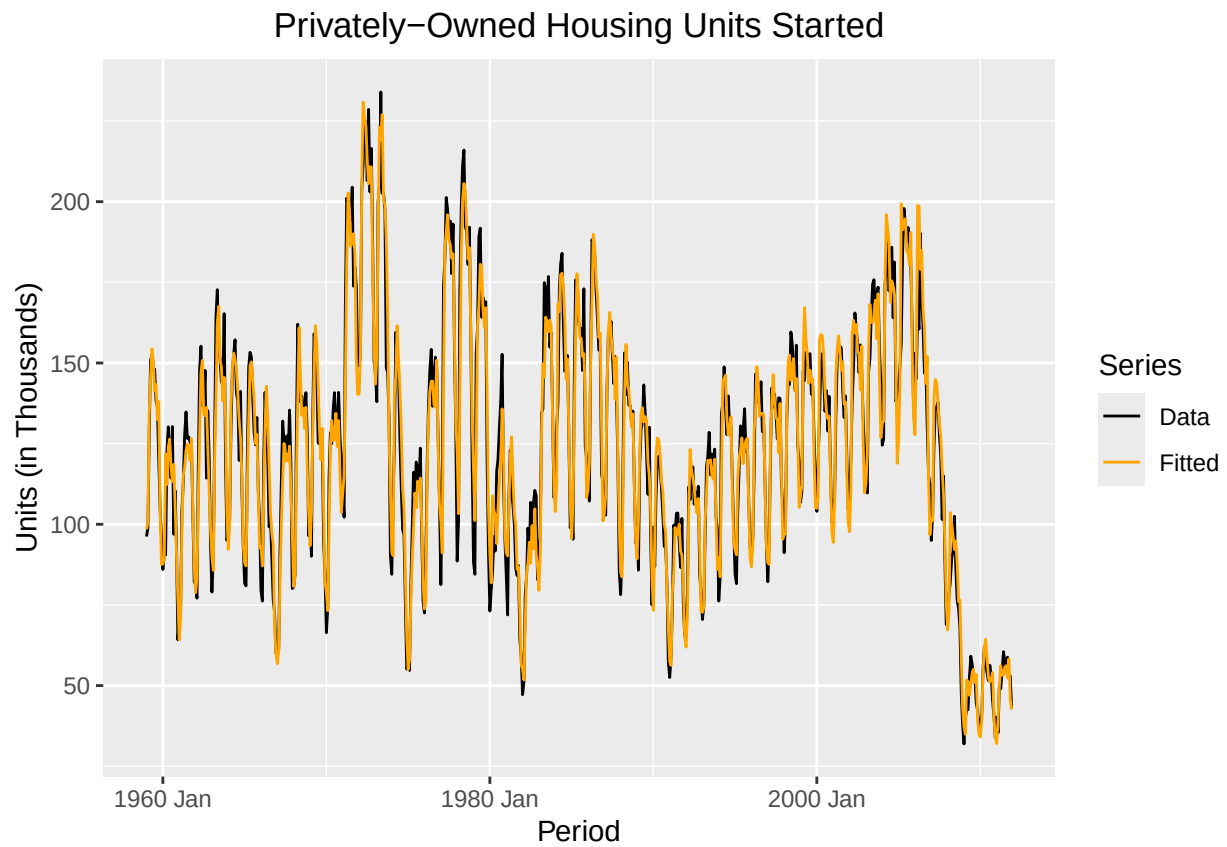


ETS ETS Fit

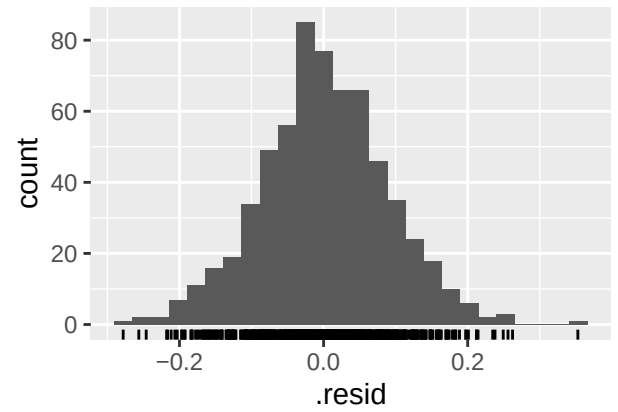
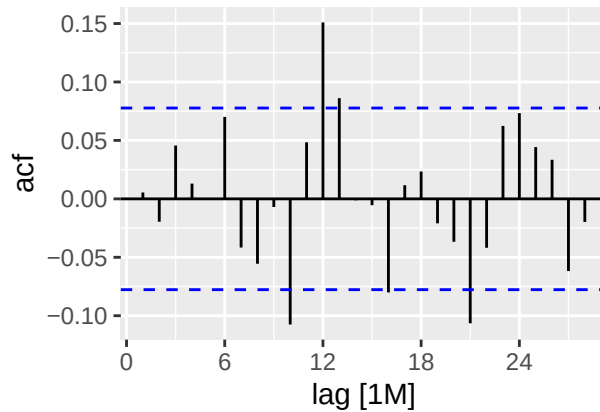
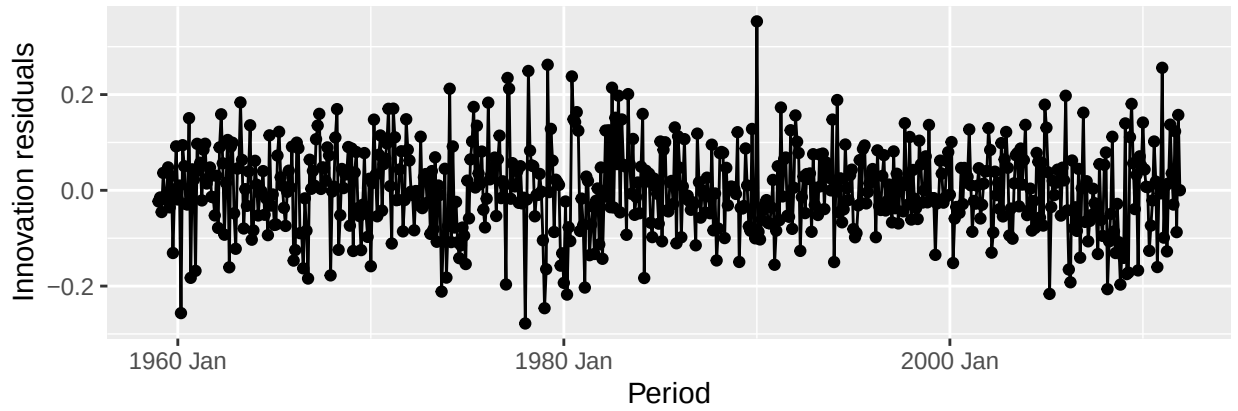
```
## Series: Value
## Model: ETS(M,Ad,M)
## Smoothing parameters:
##   alpha = 0.6932812
##   beta  = 0.0001518543
##   gamma = 0.0001647222
##   phi   = 0.9771635
##
## Initial states:
##   l[0]      b[0]      s[0]      s[-1]  s[-2]      s[-3]      s[-4]      s[-5]
## 137.927 -0.7052112 0.7590073 0.9028882 1.0795 1.052248 1.105566 1.121532
##   s[-6]      s[-7]      s[-8]      s[-9]      s[-10]      s[-11]
## 1.174574 1.186762 1.141264 1.011063 0.7480388 0.7175573
##
## sigma^2: 0.0082
##
##      AIC      AICc      BIC
## 7122.391 7123.500 7202.585
```

ETS(Value)

<ETS(M,Ad,M)>



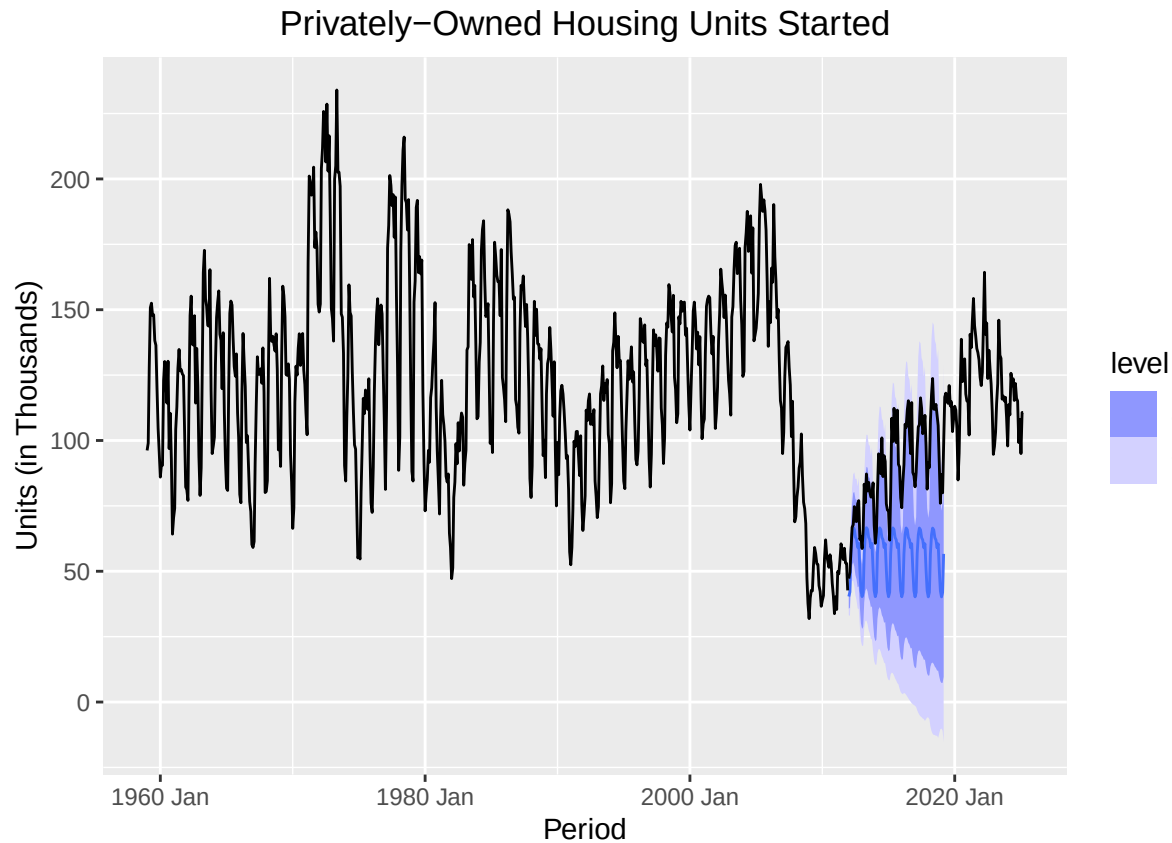
The ETS function selected a multiplicative error, additive damped trend, and multiplicative seasonal component. Beta is close to zero which corresponds to the lack of a trend in the dataset.



Residuals

.model	lb_stat	lb_pvalue
ETS(Value)	57.30947	0.0001516

The residuals are normally distributed around zero. There are clear spikes in the residuals but no overall trend. Prediction intervals may be robust. The autocorrelation of the residuals however does not resemble white noise. The Ljung-Box test clearly shows significance. The model does not include all relevant information and can be improved.



Test Set Evaluation

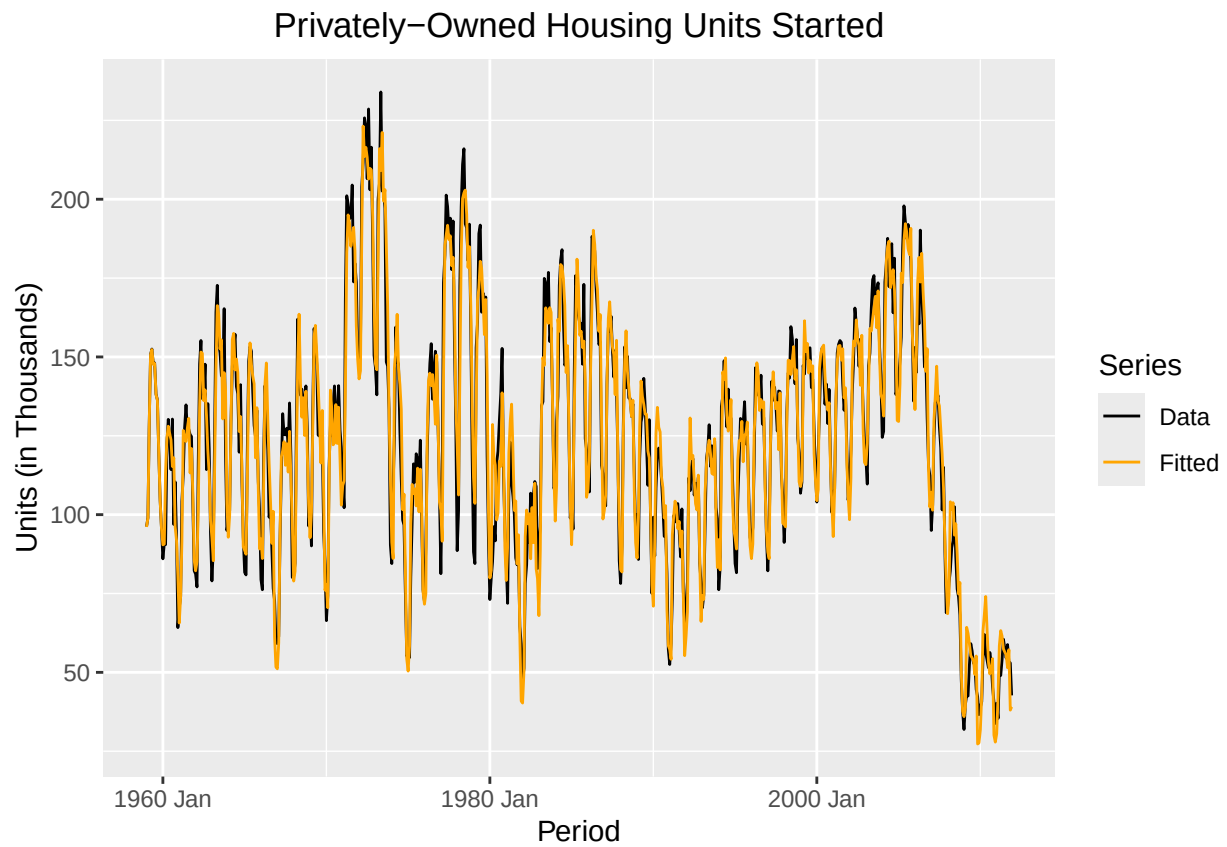
The model is very inaccurate. It fails to see the increase in housing starts. This is likely because it values more recent observations. Starting at the recession low points thus biases the predictions.

ARIMA

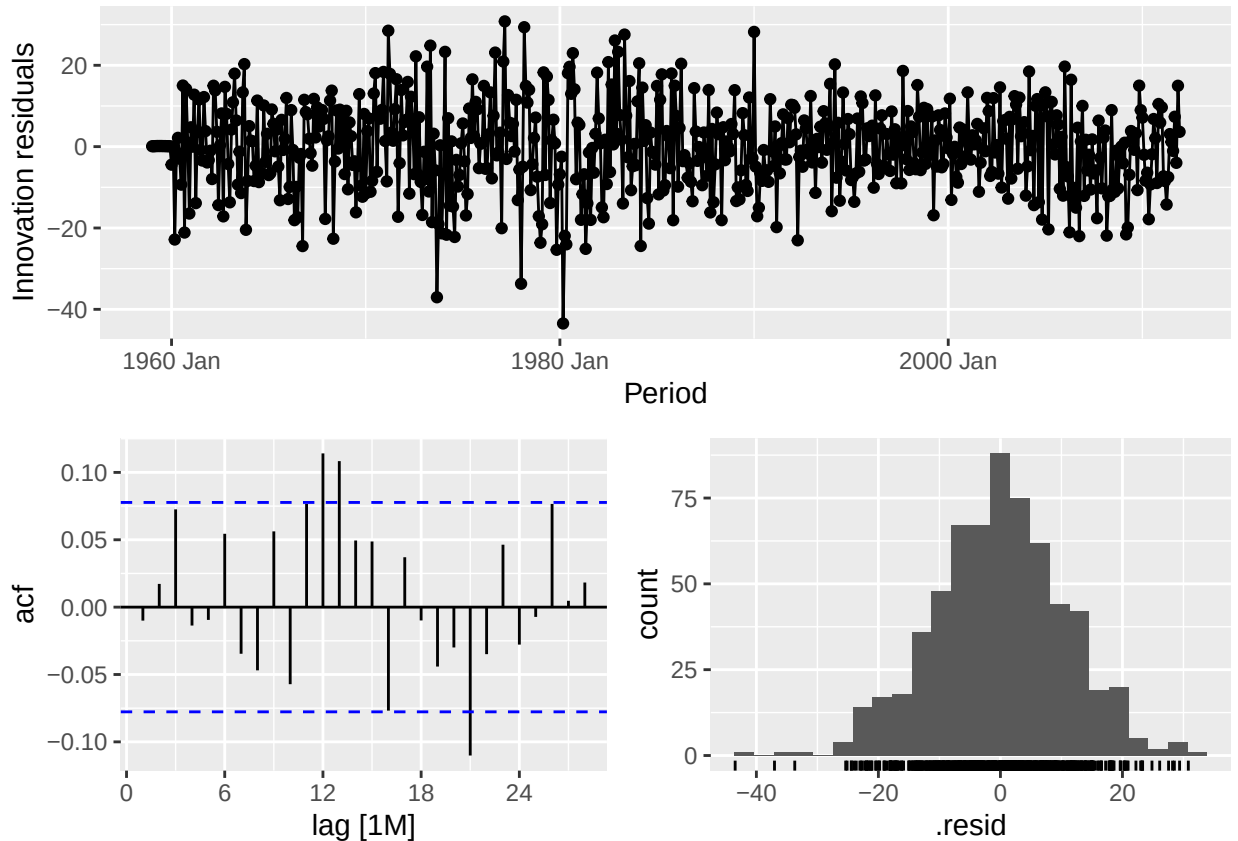
ARIMA Fit

```
## Series: Value
## Model: ARIMA(1,0,1)(0,1,1)[12]
##
## Coefficients:
##      ar1      ma1      sma1
##      0.9731 -0.2885 -0.8652
## s.e.  0.0099  0.0384  0.0254
##
## sigma^2 estimated as 117.8:  log likelihood=-2380.45
## AIC=4768.91  AICc=4768.97  BIC=4786.65
```

```
ARIMA(Value, stepwise = FALSE, approximation = FALSE)
<ARIMA(1,0,1)(0,1,1)[12]>
```



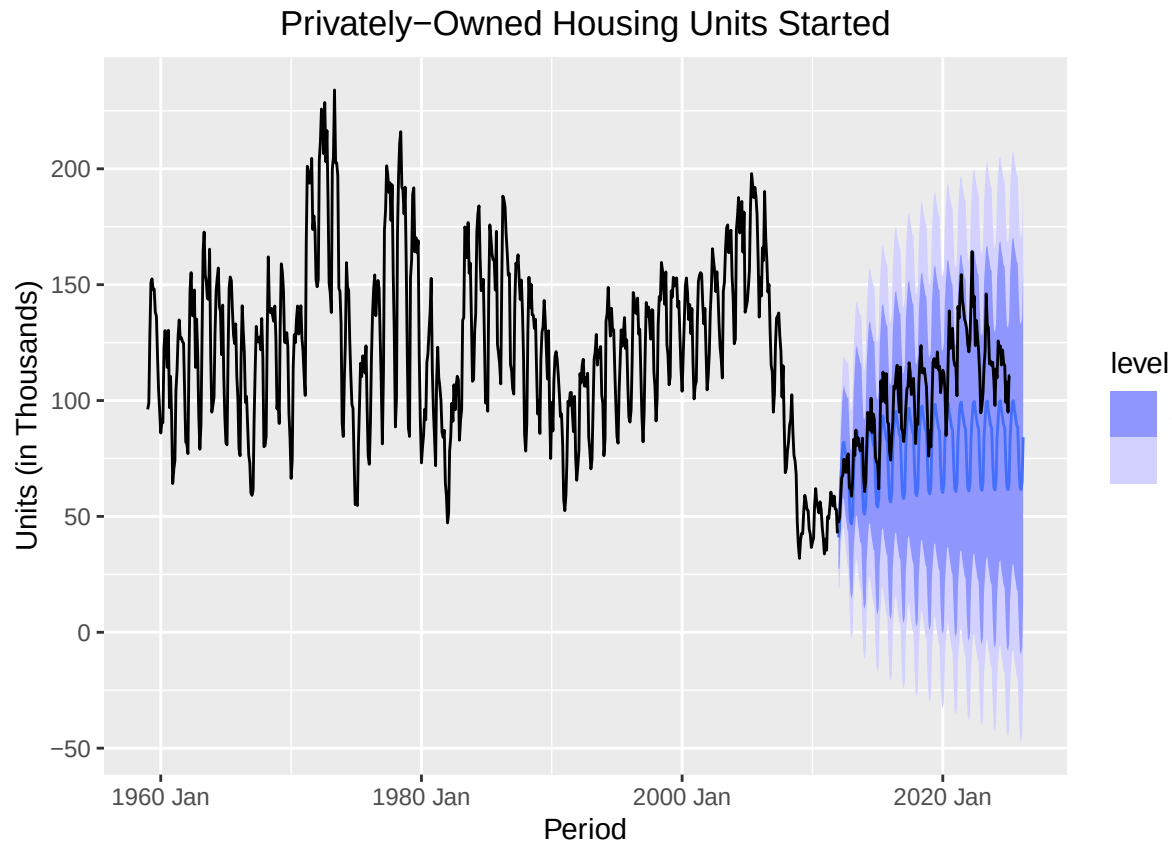
The ARIMA model includes a seasonal component. The seasonal component does not include auto regression. It is seasonally differenced.



Residuals

.model	lb_stat	lb_pvalue
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	52.57546	0.0001582

The residuals are normally distributed around zero. There are clear spikes in the residuals but no overall trend. Prediction intervals may be robust. The autocorrelation of the residuals however does not resemble white noise. The Ljung-Box test clearly shows significance. The model does not include all relevant information and can be improved.



Test Set Evaluation

ARIMA somewhat predicts the recovery but it still underperforms. It could not predict the extent of the increase.

Regression with ARIMA Errors

Regression Fit

```
##
## =====
##               Model 1      Model 2      Model 3      Model 4
## -----
```

## ar1	0.97 ***	0.98 ***	1.30	1.28
##	(0.01)	(0.01)	(1.06)	(0.72)
## ma1	-0.31 ***	-0.34 ***	-0.83	-0.81
##	(0.04)	(0.04)	(1.06)	(0.71)
## sma1	-0.86 ***	-0.86 ***	-0.83 ***	-0.83 ***
##	(0.02)	(0.03)	(0.06)	(0.06)
## UNRATE	-4.73 *		-4.75	
##	(2.01)		(2.89)	
## FEDFUNDS	-1.68 *		-2.22	
##	(0.68)		(1.69)	
## PPI_Construction	-0.21		-0.39	
##	(0.30)		(0.33)	
## UNRATE_Lag		-2.62		-0.57
##		(1.99)		(2.90)
## FEDFUNDS_Lag		-3.31 ***		-1.50
##		(0.66)		(1.74)

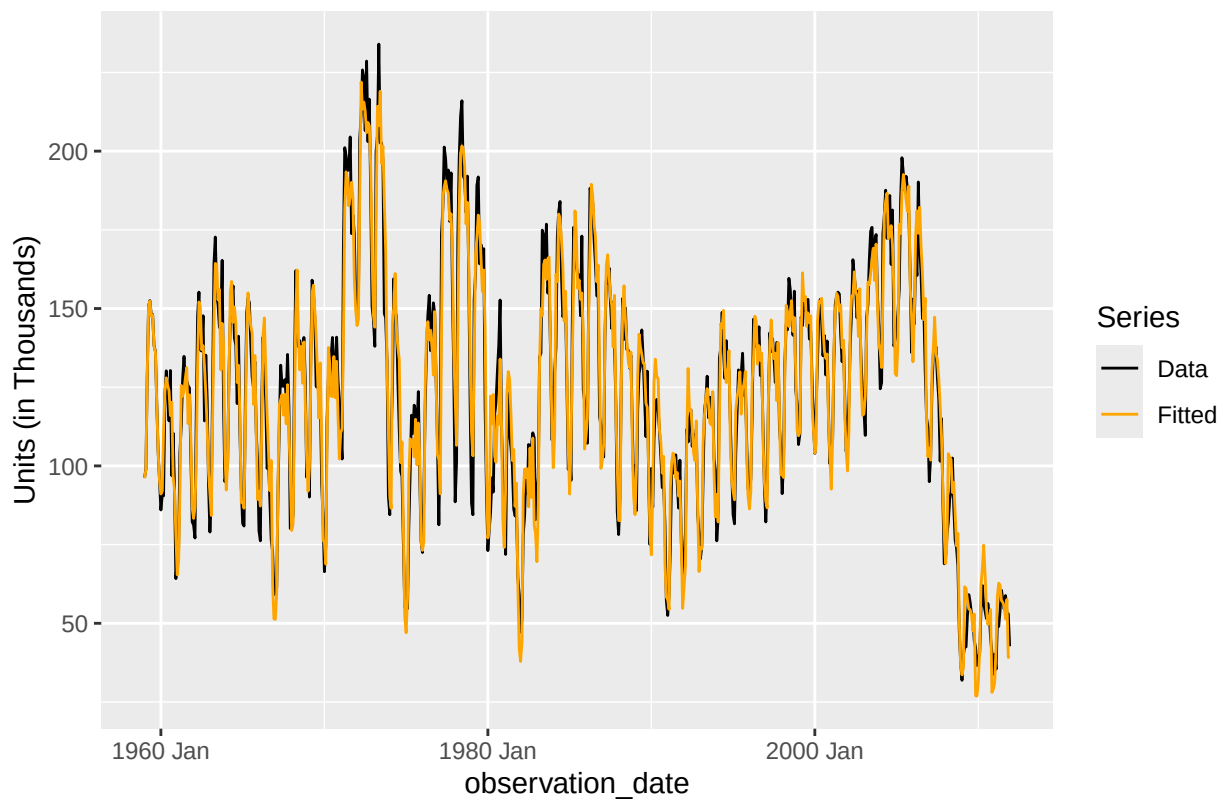
```

## PPI_Construction_Lag          -0.47          -0.68 *
##                               (0.31)          (0.33)
## ar2                          -0.31          -0.29
##                               (1.05)          (0.71)
## ma2                           0.18          0.19
##                               (0.55)          (0.35)
## sar1                           0.12          0.14
##                               (0.08)          (0.08)
## Case.Shiller                   0.73          0.92 *
##                               (0.44)          (0.44)
## -----
## sigma2           116.73           113.50           75.84           75.31
## log_lik          -2375.95          -2367.05          -1032.76          -1031.63
## AIC              4765.89           4748.11           2087.51           2085.26
## AICc             4766.07           4748.29           2088.47           2086.22
## BIC              4796.94           4779.16           2127.80           2125.55
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05

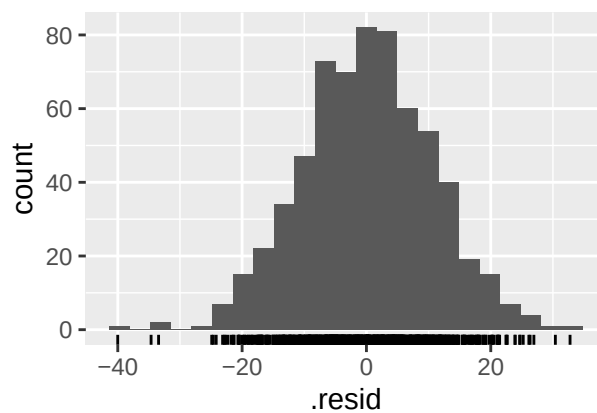
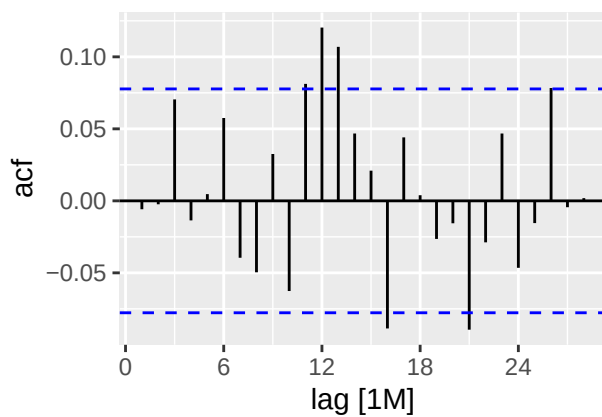
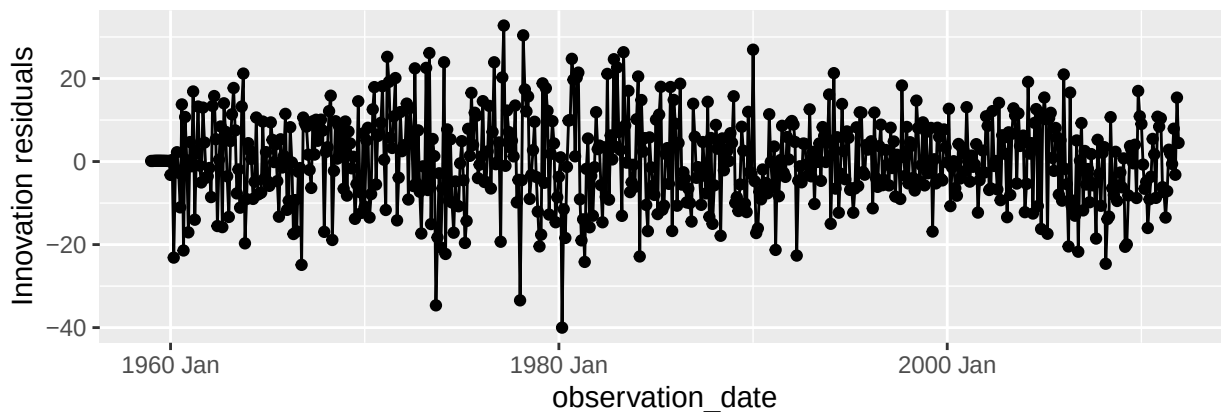
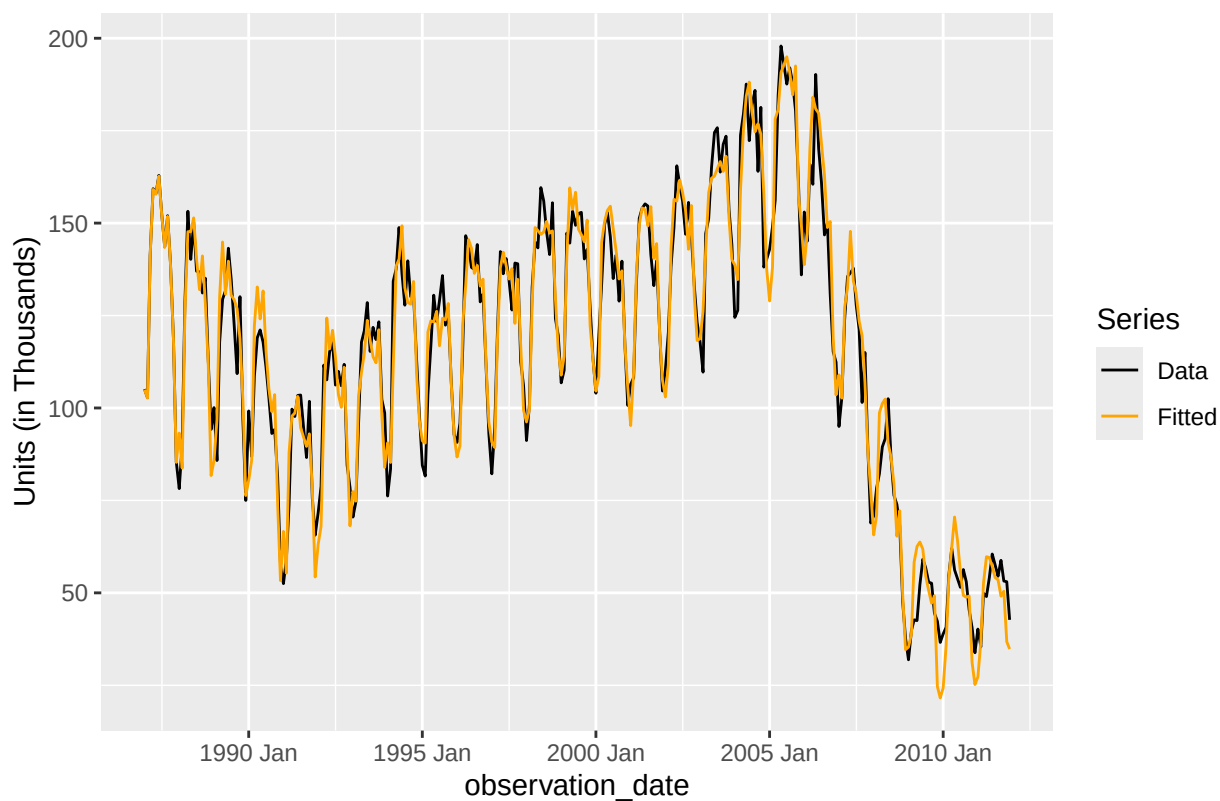
```

The best models were the regressions with lag. The best overall model was the regression with Case-Shiller.

Privately-Owned Housing Units Started No Case-Shiller

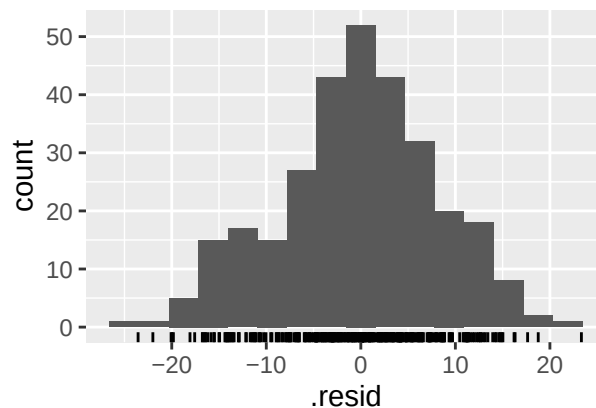
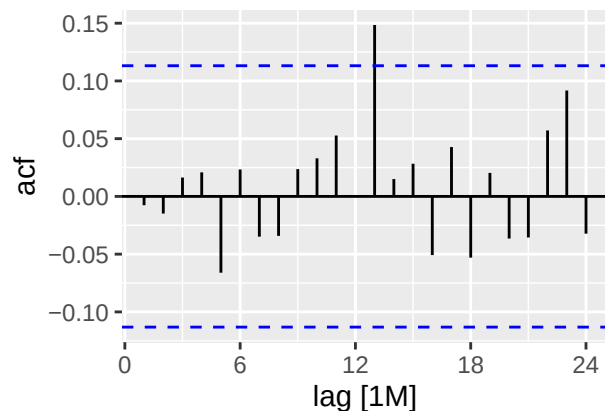
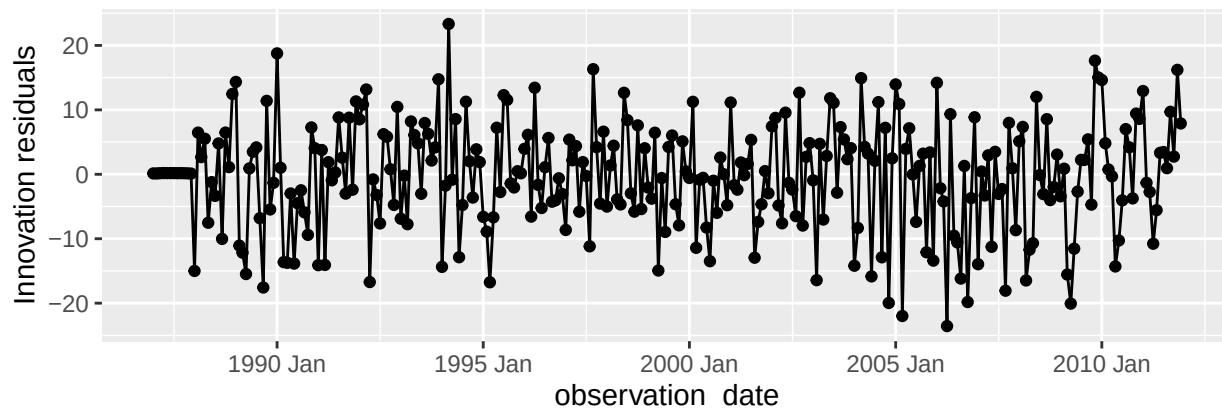


Privately-Owned Housing Units Started w/ Case-Shiller



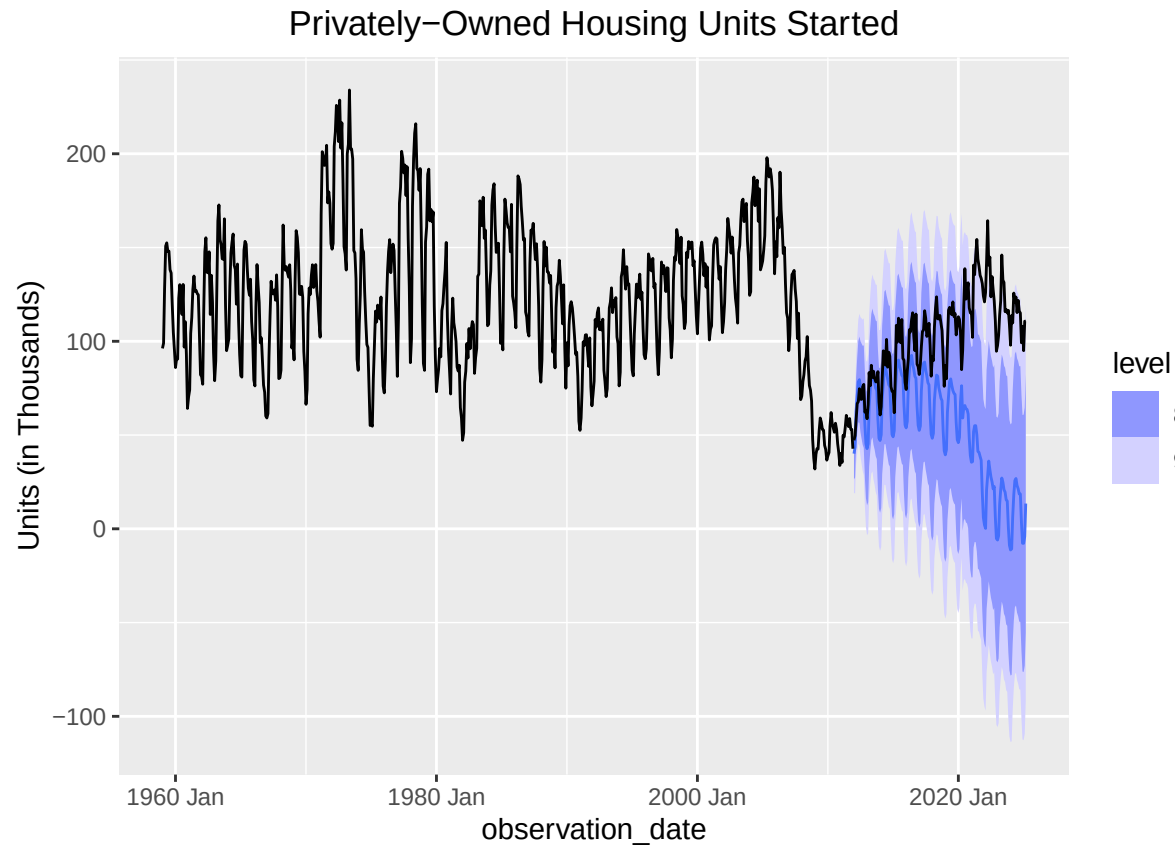
Residuals

.model	lb_stat	lb_pvalue
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	49.89722	0.000377

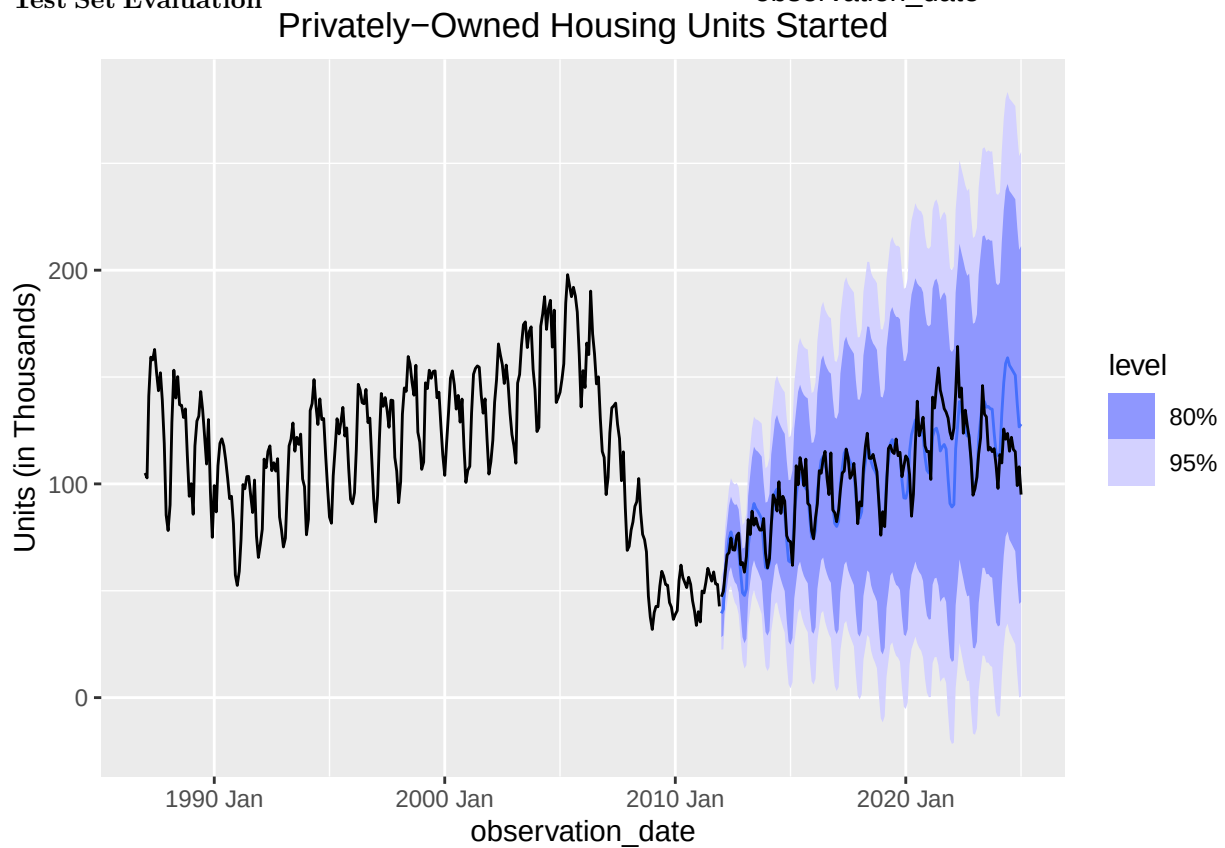


.model	lb_stat	lb_pvalue
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + Case.Shiller + pdq())	18.65175	0.6074691

Both regression have normally distributed residuals around zero. There are clear spikes in the residuals but no overall trend for the regression without Case-Shiller. There is a sudden spike in residuals in the second regression. Unlike the regression without Case-Shiller, the second regression's ACF does resemble white noise. The Ljung-Box test shows no significance.



Test Set Evaluation



The regression without Case-Shiller starts accurately predicting the outcomes but begins to show a decrease

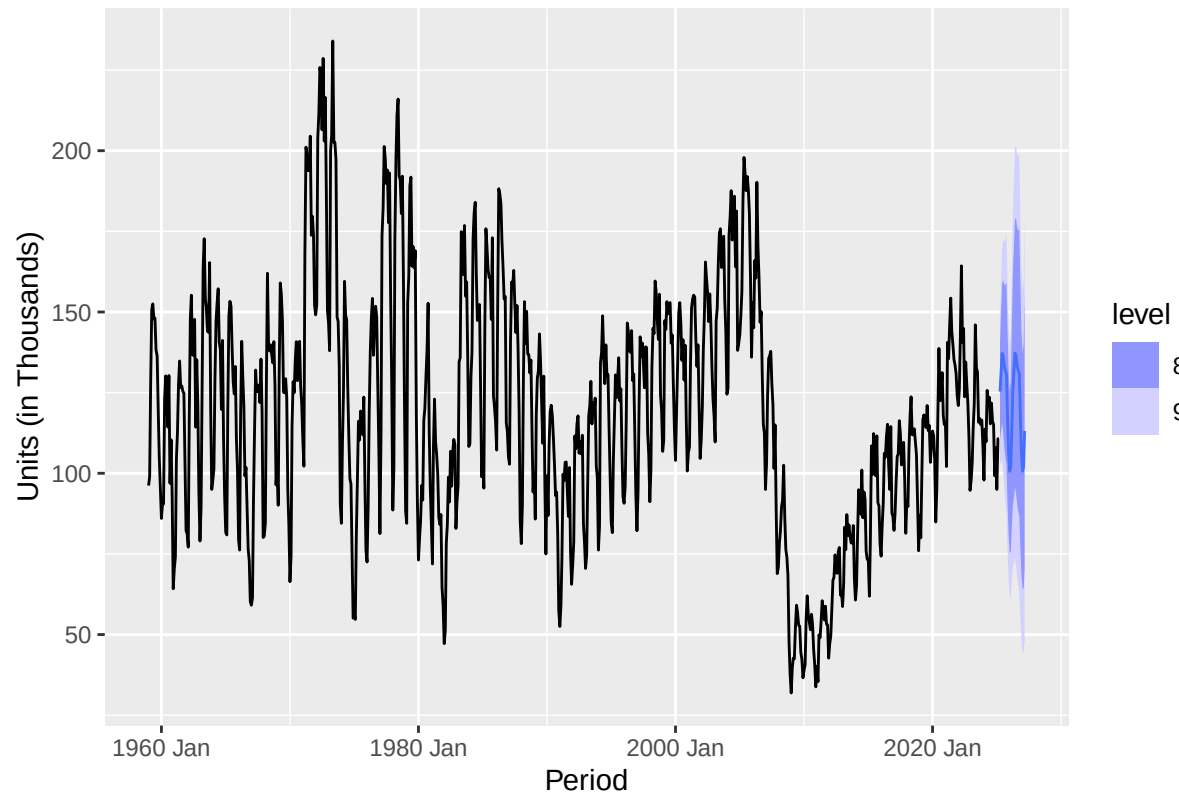
when they should be a increase. The second regression is much more accurate.

Model Accuracy

.model	.type	RMSE	MAE	MAPE	MASE	RMSSE
ETS(Value)	Training	10.579827	8.162330	7.053143	0.413988	0.4049727
ETS(Value)	Test	51.515038	47.099064	3.601504	2.388833	2.9718833
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	Training	10.727053	8.440357	7.635093	0.428089	0.4106082
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	Test	29.511124	24.954802	2.836931	2.265691	0.1296214
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	Training	10.501903	8.320612	7.538949	0.422015	0.4019900
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	Test	64.808913	51.215967	5.558484	2.597639	2.4807438
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + Case.Shiller + pdq())	Training	8.353941	6.551291	6.802651	0.449531	0.4378054
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + Case.Shiller + pdq())	Test	14.126209	9.920718	9.388045	0.503172	0.5407201

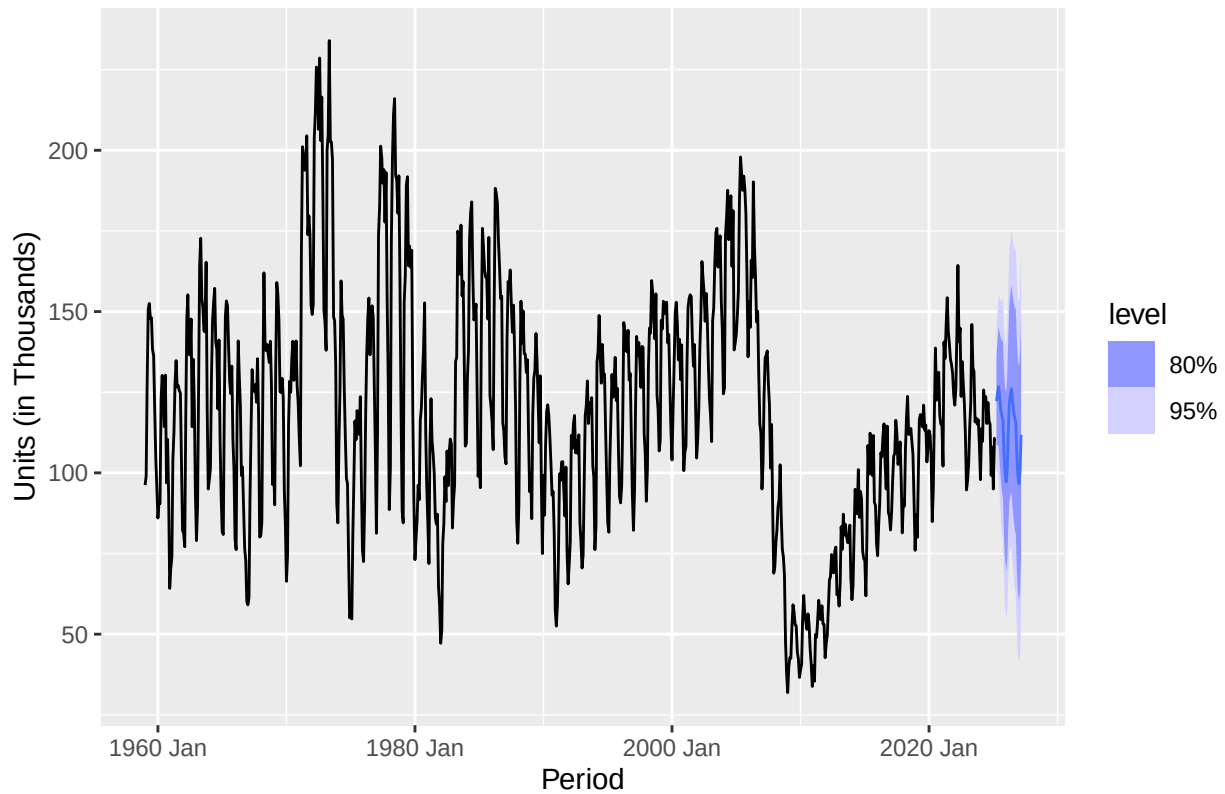
The best model is the regression with lagged predictors and Case-Shiller. The second best model is ARIMA. Interestingly, the ARIMA and regression without Case-Shiller have similar training errors but quite different test set errors. ARIMA performs better compared to the regression.

Privately-Owned Housing Units Started ETS 2 Yrs Forecast

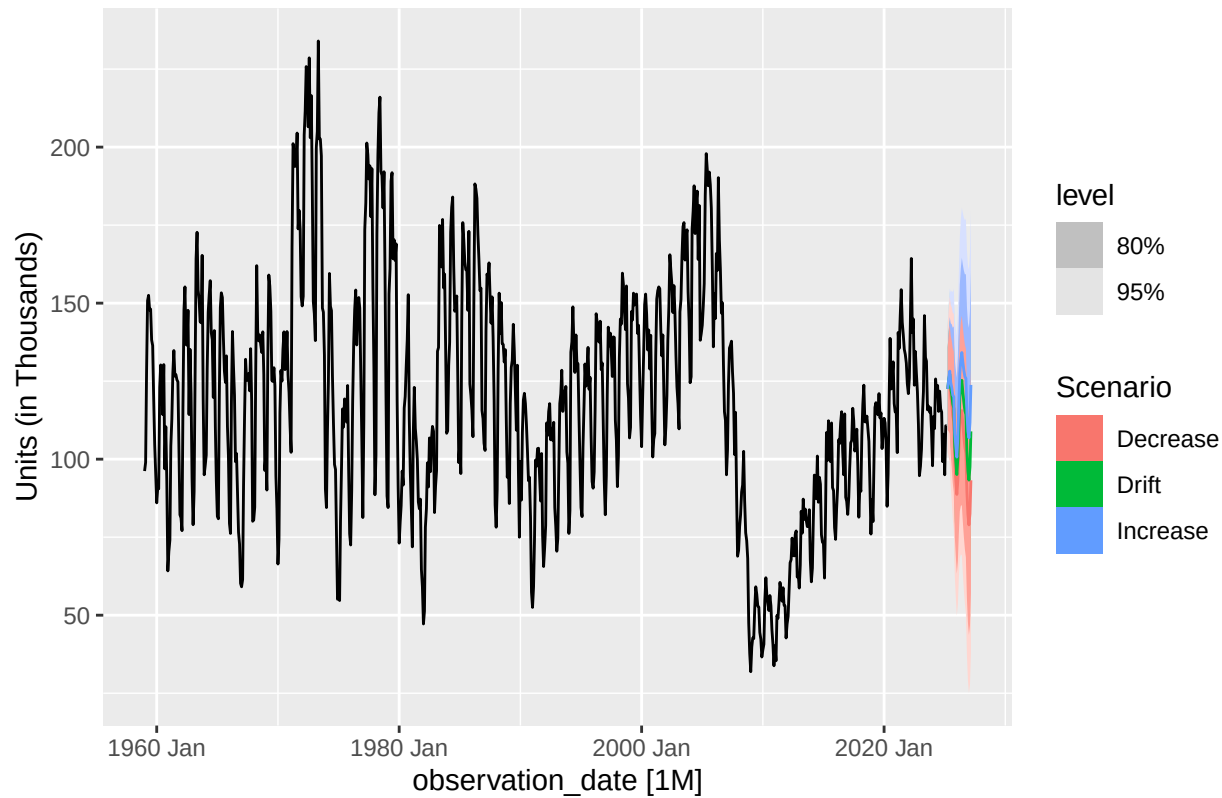


Two Year Forecast

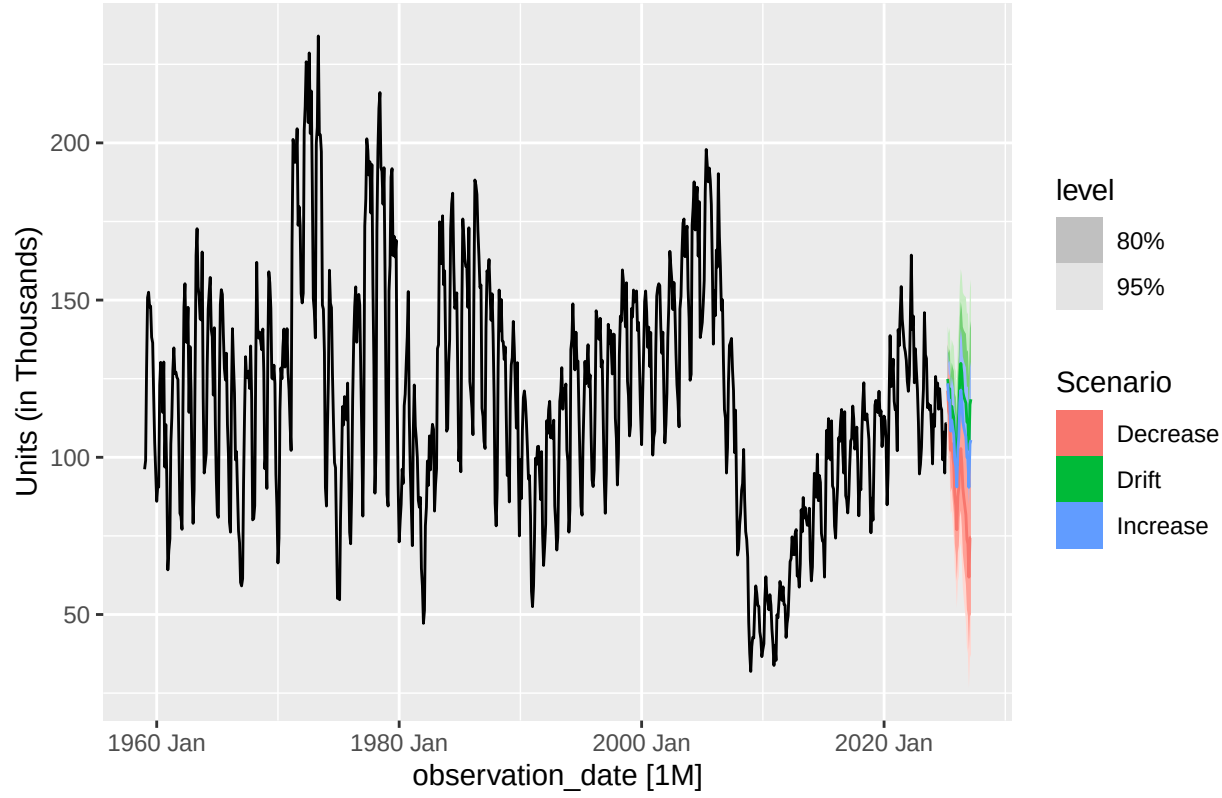
Privately-Owned Housing Units Started ARIMA 2 Yrs Forecast



Privately-Owned Housing Units Started Regression1 1 Yr Forecast



Privately-Owned Housing Units Started Regression2 1 Yr Forecast



90/10 Training/Test Set

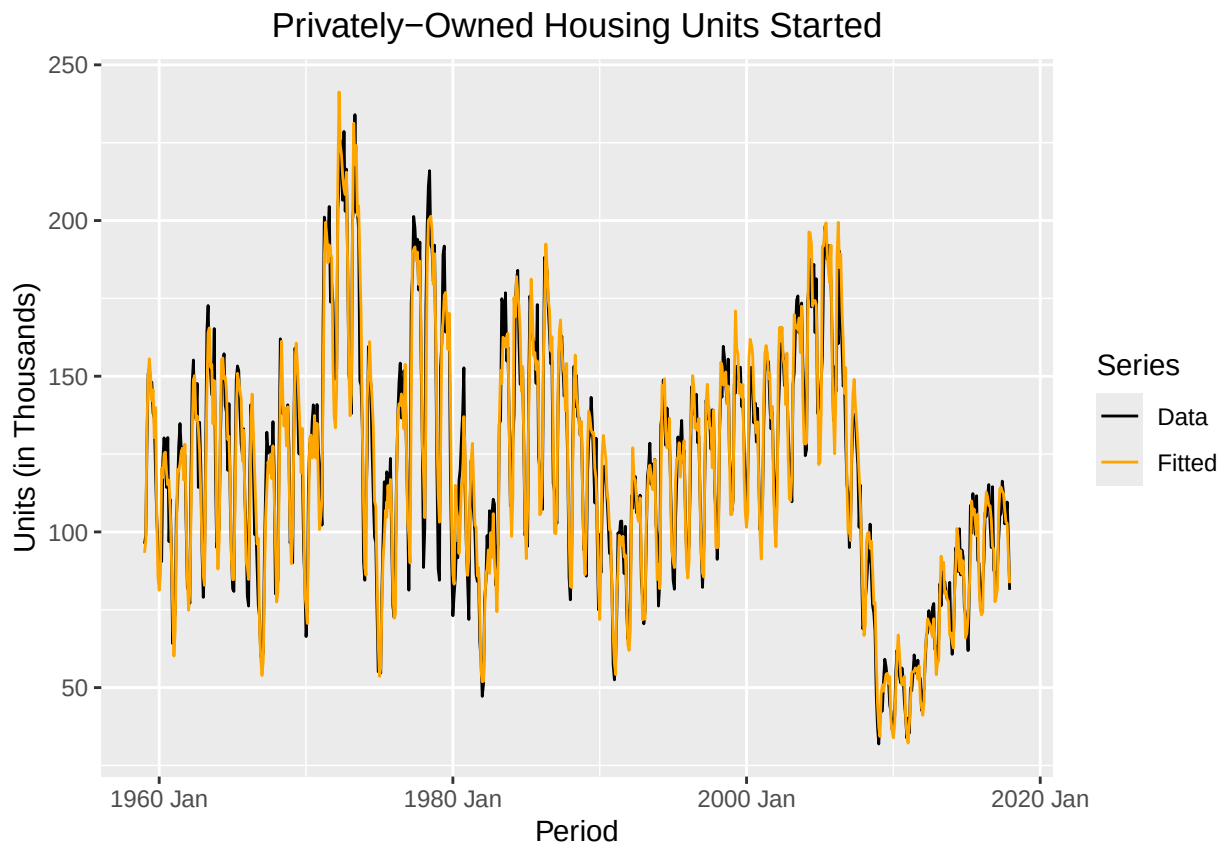
ETS

ETS Fit

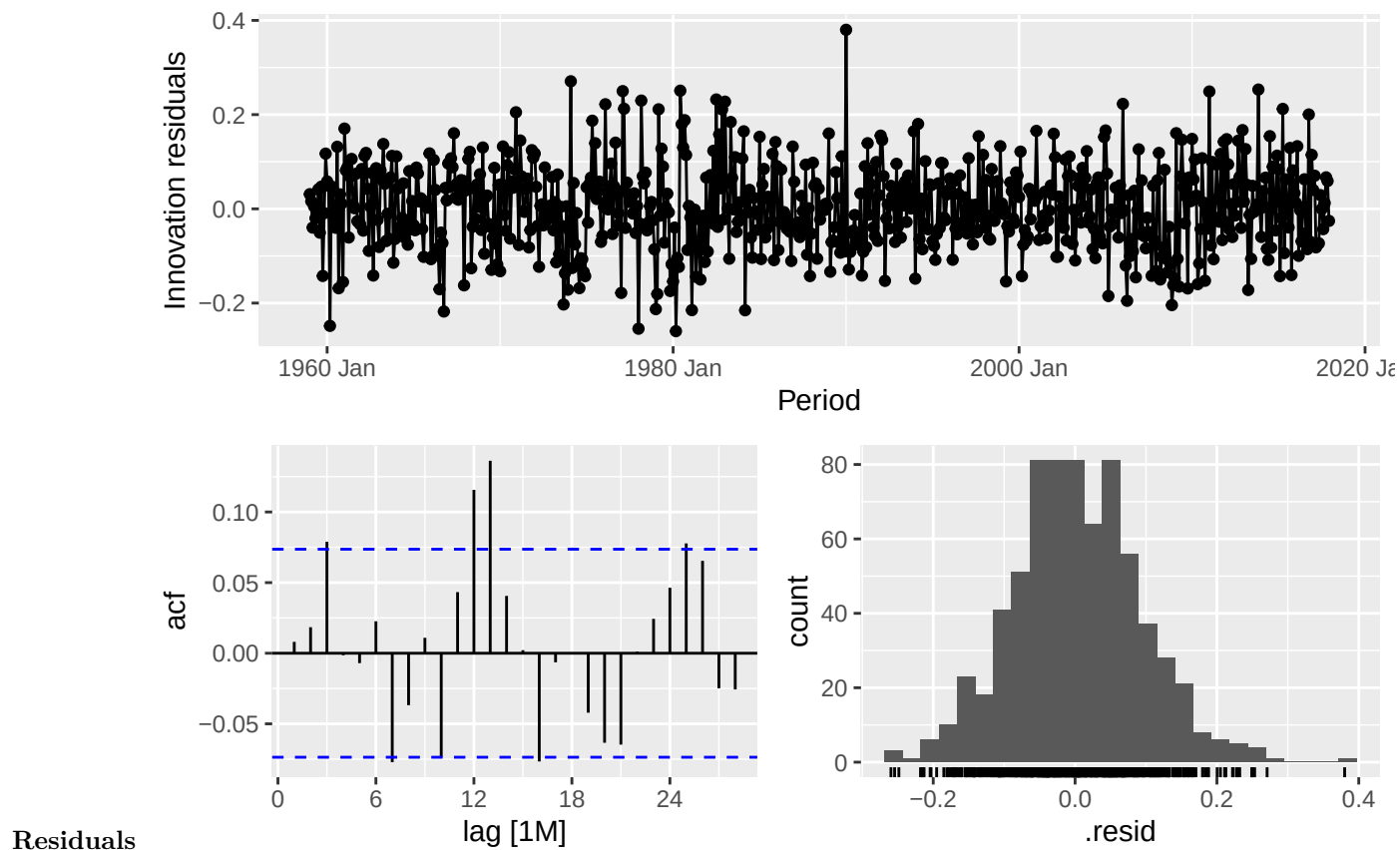
```
## Series: Value
## Model: ETS(M,Ad,M)
## Smoothing parameters:
##   alpha = 0.6487965
##   beta  = 0.000101315
##   gamma = 0.05744239
##   phi   = 0.970119
##
## Initial states:
##   l[0]    b[0]    s[0]    s[-1]    s[-2]    s[-3]    s[-4]    s[-5]
## 137.5345 -1.561529 0.7716042 0.9379711 1.116233 1.061147 1.135767 1.12865
##   s[-6]    s[-7]    s[-8]    s[-9]    s[-10]    s[-11]
## 1.167325 1.183585 1.133592 0.9685729 0.7096654 0.6858863
##
## sigma^2: 0.0084
##
##      AIC      AICc      BIC
## 7967.731 7968.723 8049.855
```

ETS(Value)

<ETS(M,Ad,M)>



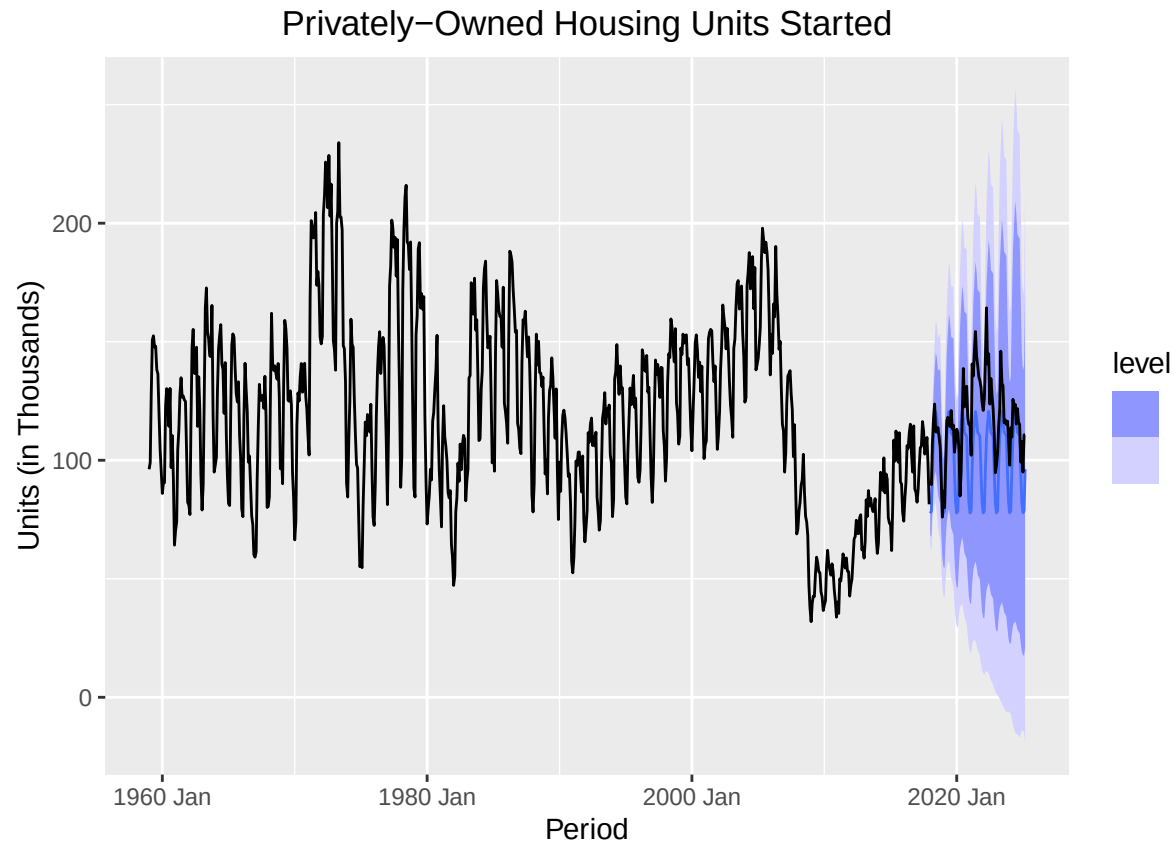
The ETS function selected a multiplicative error, additive damped trend, and multiplicative seasonal component. Beta is close to zero which corresponds to the lack of a trend in the dataset. The gamma is larger than in the 80/20 split. The model weighs more recent observation seasonality more.



Residuals

.model	lb_stat	lb_pvalue
ETS(Value)	53.56618	0.0004868

The residuals are normally distributed around zero. There are clear spikes in the residuals but no overall trend. Prediction intervals may be robust. The autocorrelation of the residuals however does not resemble white noise. The Ljung-Box test clearly shows significance. The model does not include all relevant information and can be improved.



Test Set Evaluation

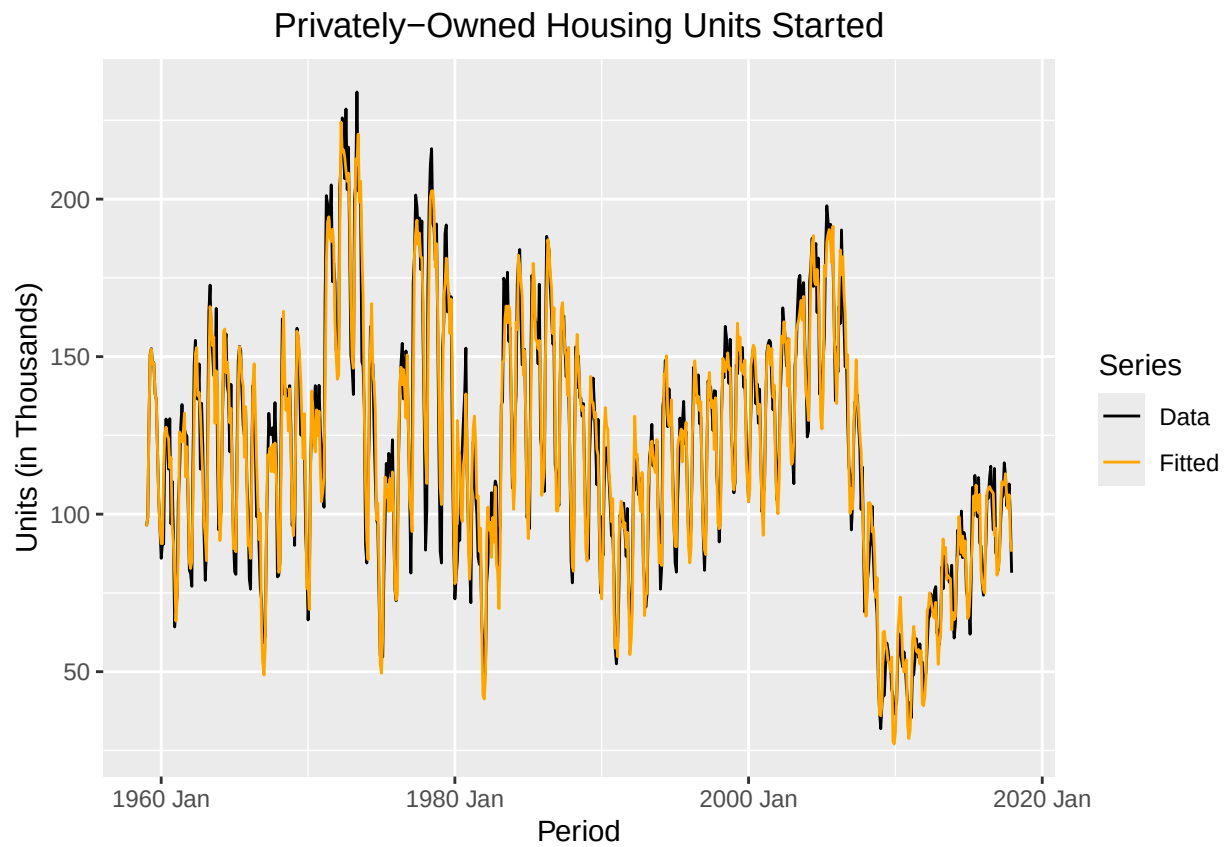
The model consistently under predicts the test set values but it performs better than the ETS model created with a 90/10 split.

ARIMA

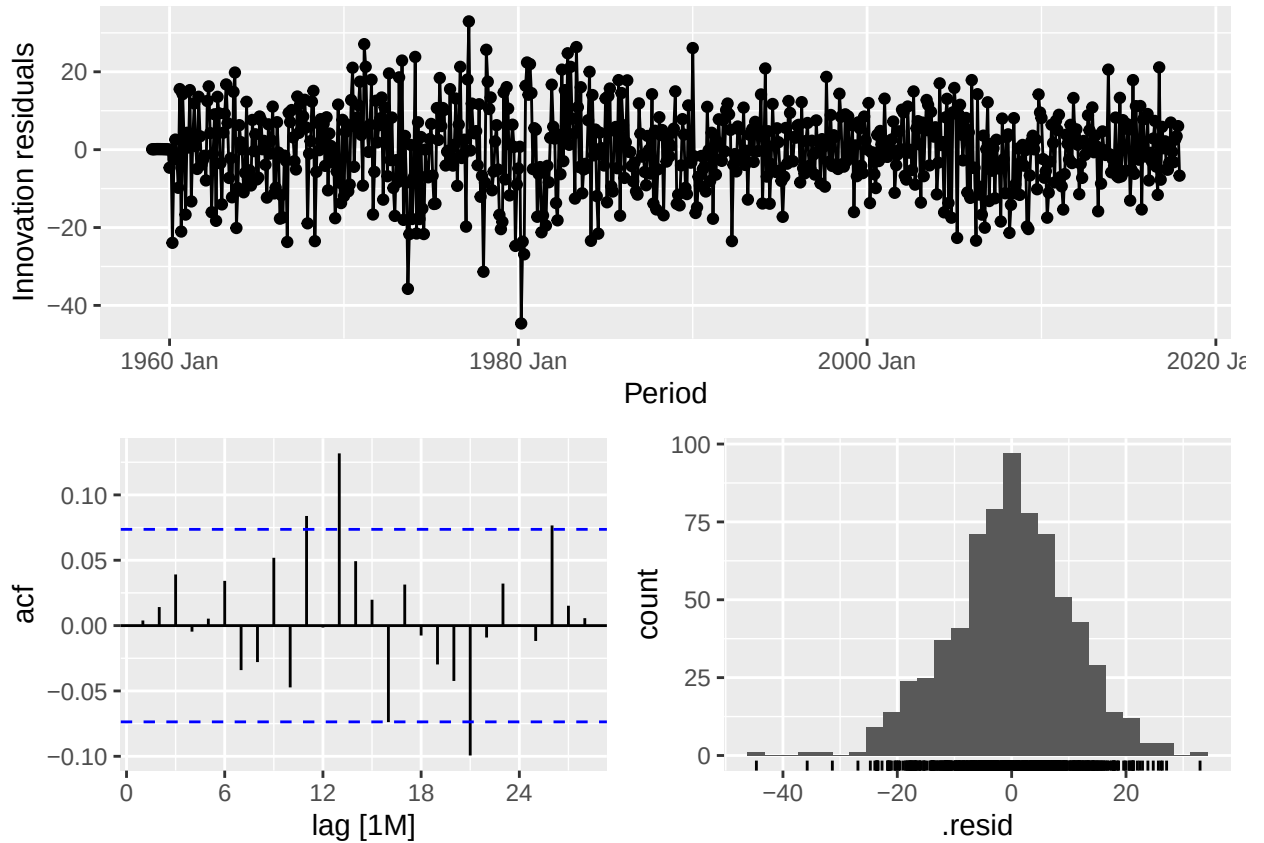
ARIMA Fit

```
## Series: Value
## Model: ARIMA(3,0,0)(1,1,1)[12]
##
## Coefficients:
##      ar1      ar2      ar3      sar1      sma1
##      0.6265  0.2406  0.0865  0.1487 -0.8874
## s.e.  0.0387  0.0440  0.0378  0.0442  0.0204
##
## sigma^2 estimated as 109.8:  log likelihood=-2628.36
## AIC=5268.71   AICc=5268.83   BIC=5295.98
```

```
ARIMA(Value, stepwise = FALSE, approximation = FALSE)
<ARIMA(3,0,0)(1,1,1)[12]>
```



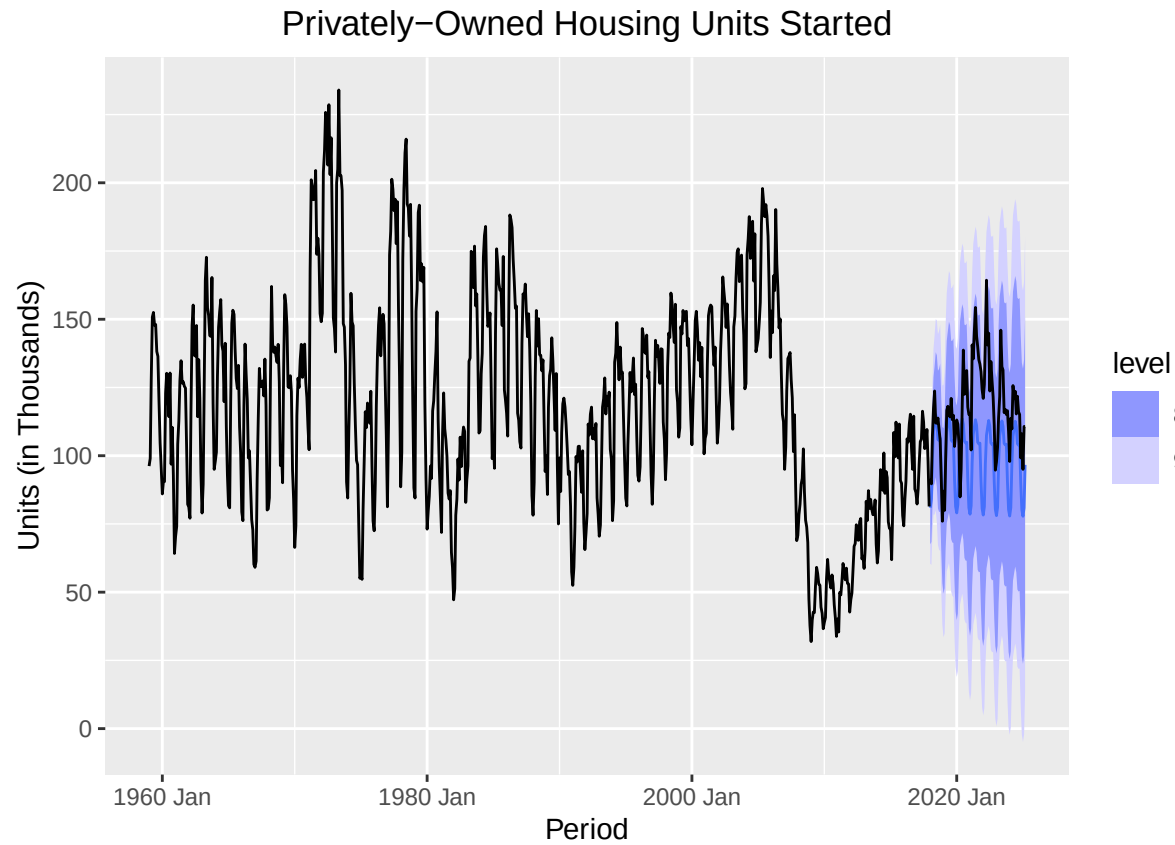
The model has seasonal differencing. Interestingly, it is a autoregression with a order of three with no moving averages in the non seasonal part of the model.



Residuals

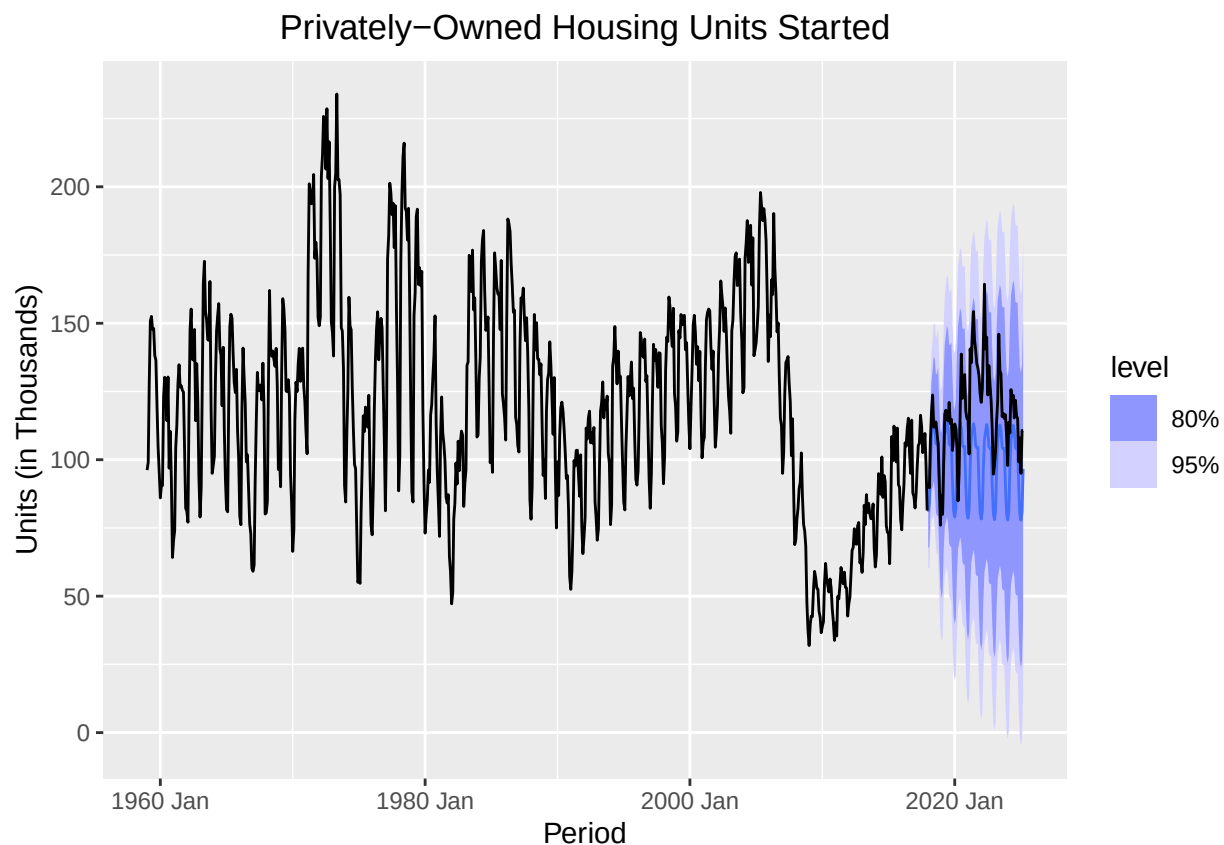
.model	lb_stat	lb_pvalue
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	41.45595	0.0049219

The residuals are normally distributed around zero. There are clear spikes in the residuals but no overall trend. Prediction intervals may be robust. The autocorrelation of the residuals however does not resemble white noise. The Ljung-Box test clearly shows significance. The model does not include all relevant information and can be improved.



Test Set Evaluation

The model is similar to the previous ARIMA regression. It also underpredicts the test values. However it performs better than the previous model.



It under predicts the test set values but it does better than the ARIMA model constructed with a 80/20 split.

Regression with ARIMA Errors

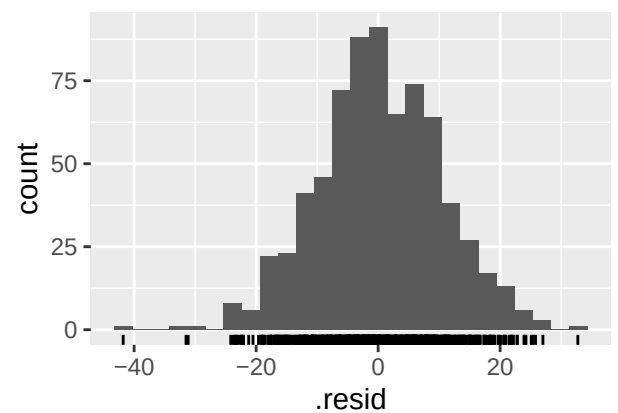
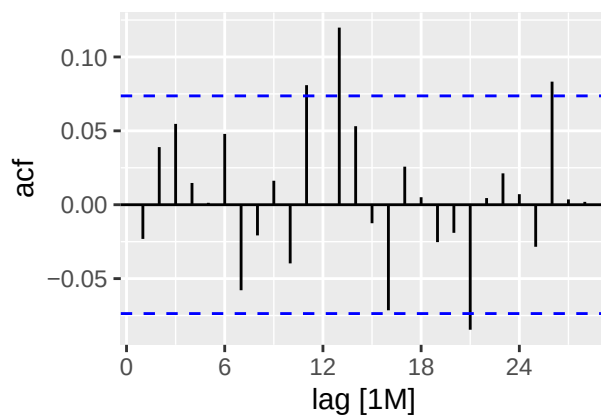
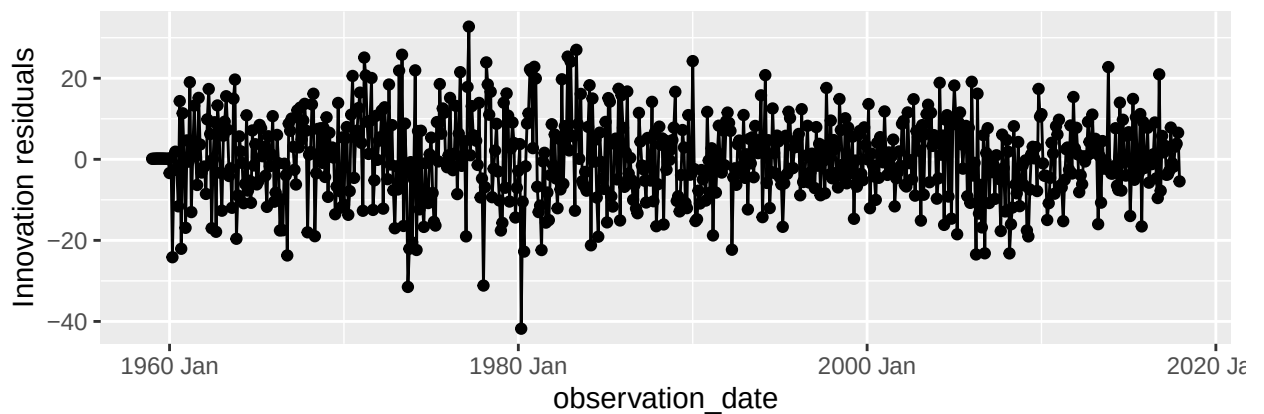
Regression Fit

```
##
## =====
##               Model 1      Model 2      Model 3      Model 4
## -----
```

## ar1	0.97 ***	0.02	1.20 *	1.17 **
##	(0.01)	(0.03)	(0.51)	(0.42)
## ma1	-0.34 ***	0.60 ***	-0.77	-0.75
##	(0.04)	(0.05)	(0.50)	(0.42)
## sar1	0.12		0.10	0.13
##	(0.20)		(0.07)	(0.07)
## sma1	-0.88 ***	-0.72 ***	-0.81 ***	-0.81 ***
##	(0.20)	(0.04)	(0.05)	(0.05)
## sma2	-0.01	-0.14 ***		
##	(0.17)	(0.04)		
## UNRATE	-4.25 *		-5.10 *	
##	(1.84)		(2.47)	
## FEDFUNDS	-1.55 *		-2.02	
##	(0.63)		(1.54)	
## PPI_Construction	-0.18		-0.33	
##	(0.25)		(0.29)	
## ar2		0.93 ***	-0.21	-0.19

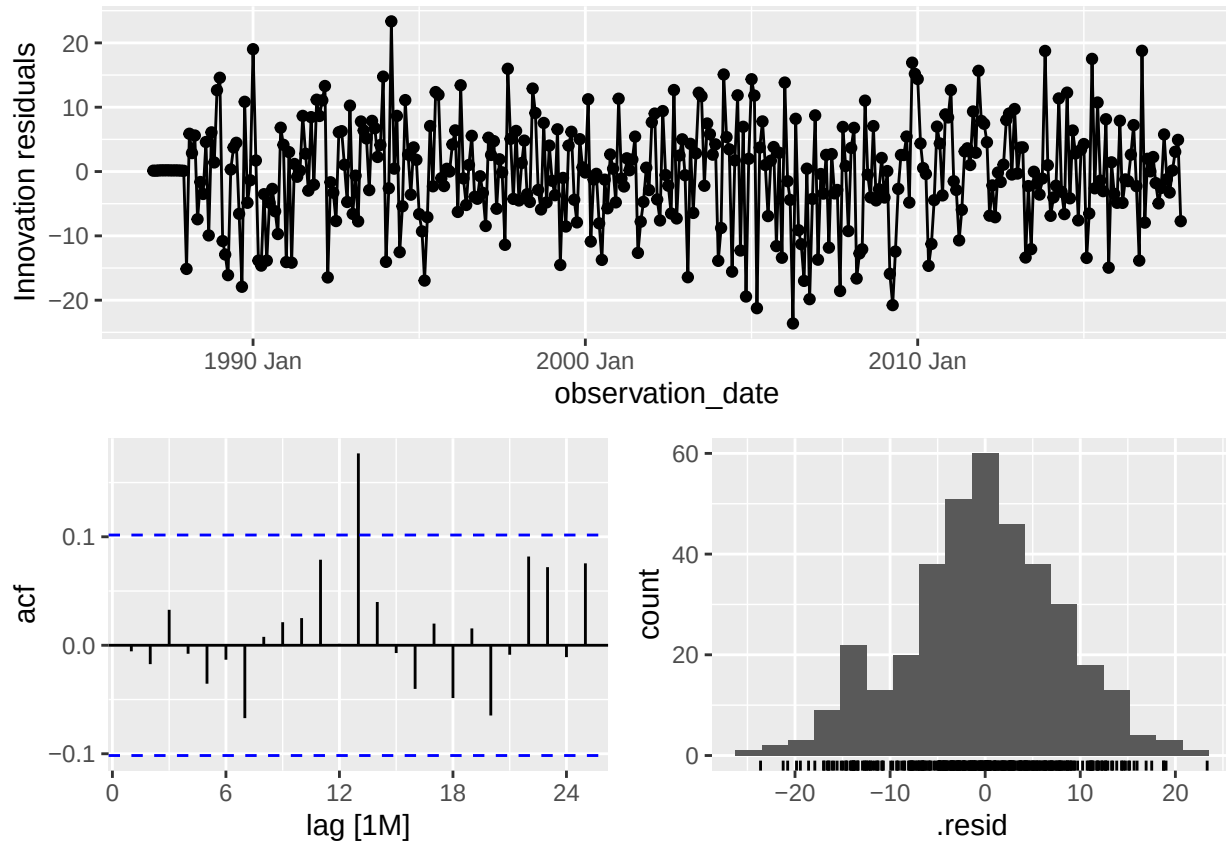
```
##              (0.03)      (0.50)      (0.42)
## ma2          -0.38 ***    0.17      0.17
##              (0.04)      (0.27)      (0.22)
## UNRATE_Lag   -2.53      (1.79)      (2.52)
## FEDFUNDS_Lag -3.30 ***    (0.61)      (1.56)
## PPI_Construction_Lag -0.42      (0.26)      (0.28)
## Case.Shiller          0.53      0.78 *
##              (0.34)      (0.33)
## -----
## sigma2        109.37      105.27      70.70      70.51
## log_lik       -2625.37     -2611.87     -1277.80     -1277.20
## AIC           5268.74      5243.73      2577.61      2576.40
## AICc          5269.00      5244.05      2578.37      2577.16
## BIC           5309.65      5289.18      2620.35      2619.15
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

The best models were the regressions with lag. The best overall model was the regression with Case-Shiller.



Residuals

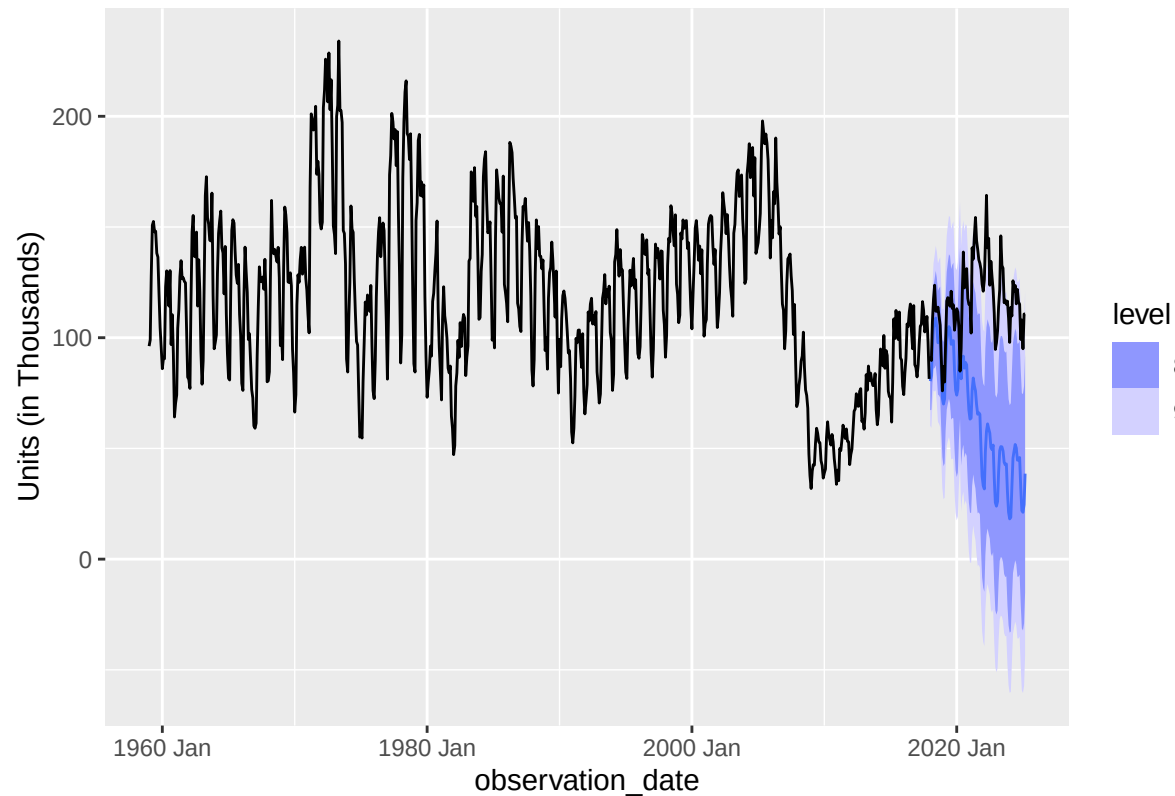
.model	lb_stat	lb_pvalue
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	37.2614	0.0157073



.model	lb_stat	lb_pvalue
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + Case.Shiller + pdq())	26.68682	0.1814614

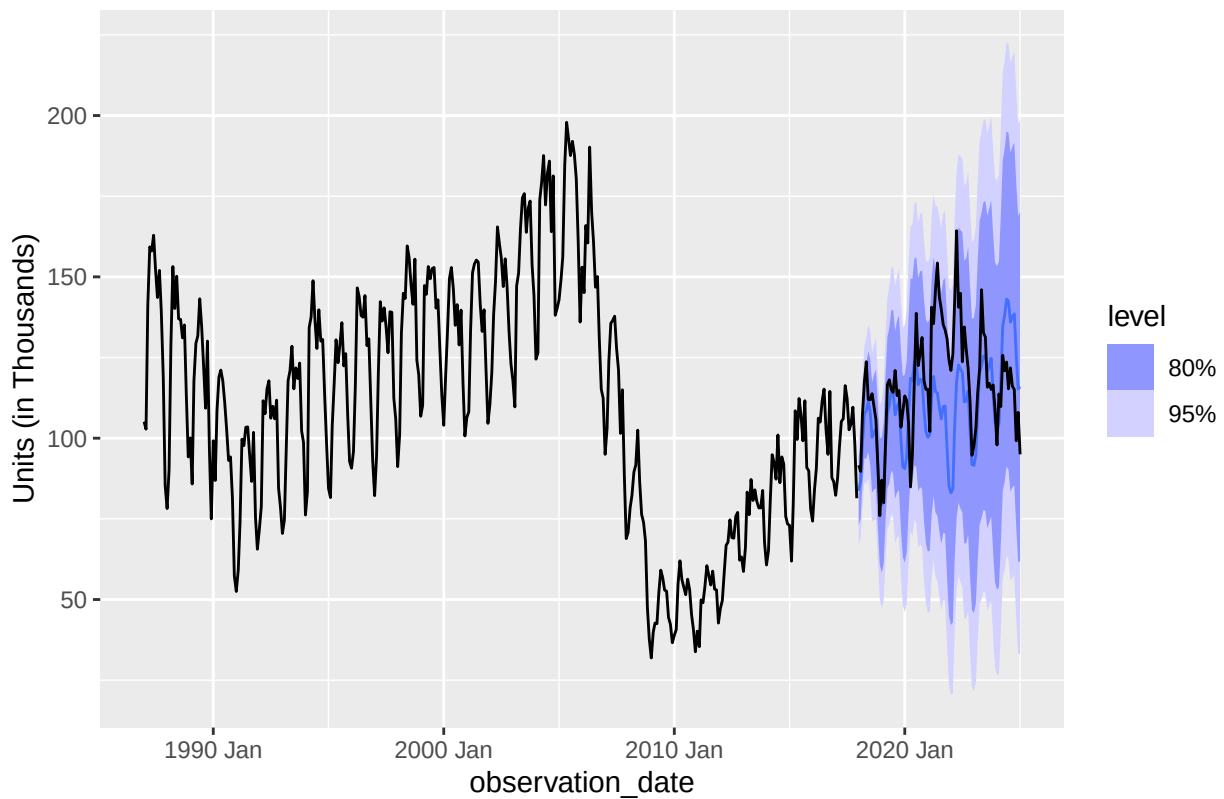
Both regression have normally distributed residuals around zero and have no overall trend. Unlike the regression without Case-Shiller, the second regression's ACF does resemble white noise. The Ljung-Box test shows no significance.

Privately-Owned Housing Units Started



Test Set Evaluation

Privately-Owned Housing Units Started



Once again, the regression with Case-Shiller performs better. The first regression predicts a downtrend which

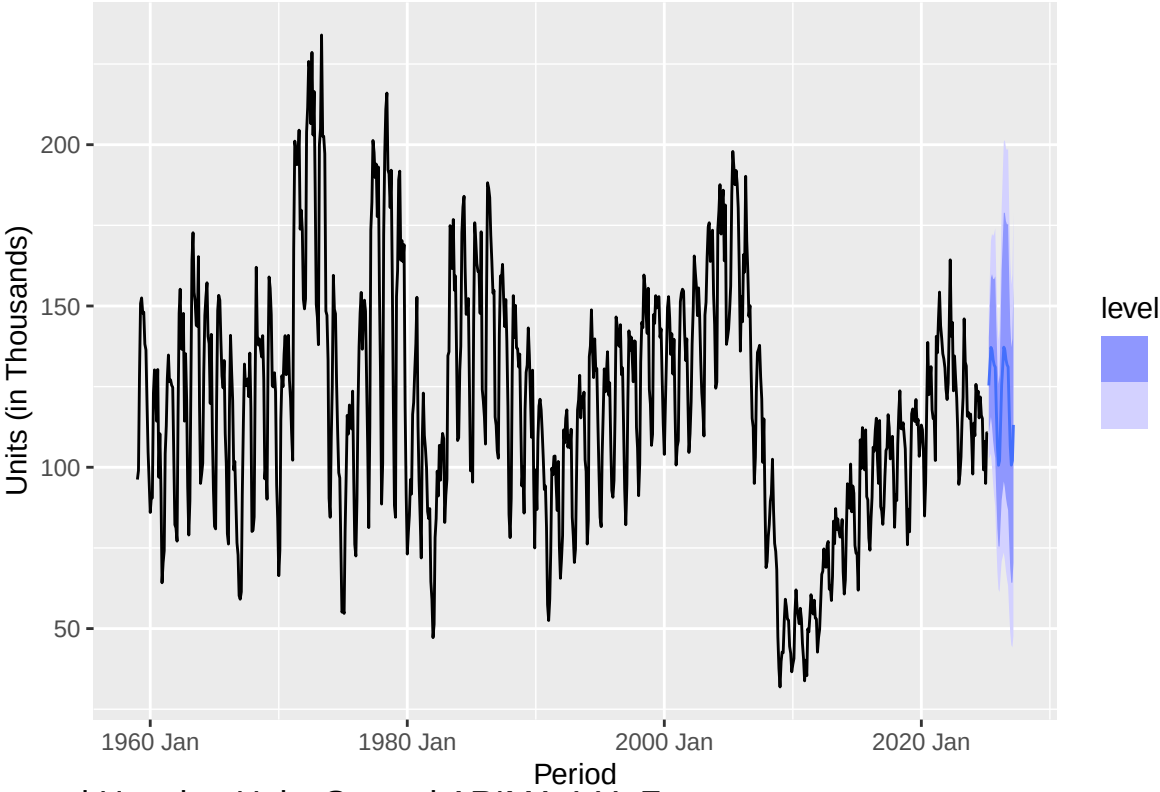
is not seen in the data.

Model Accuracy

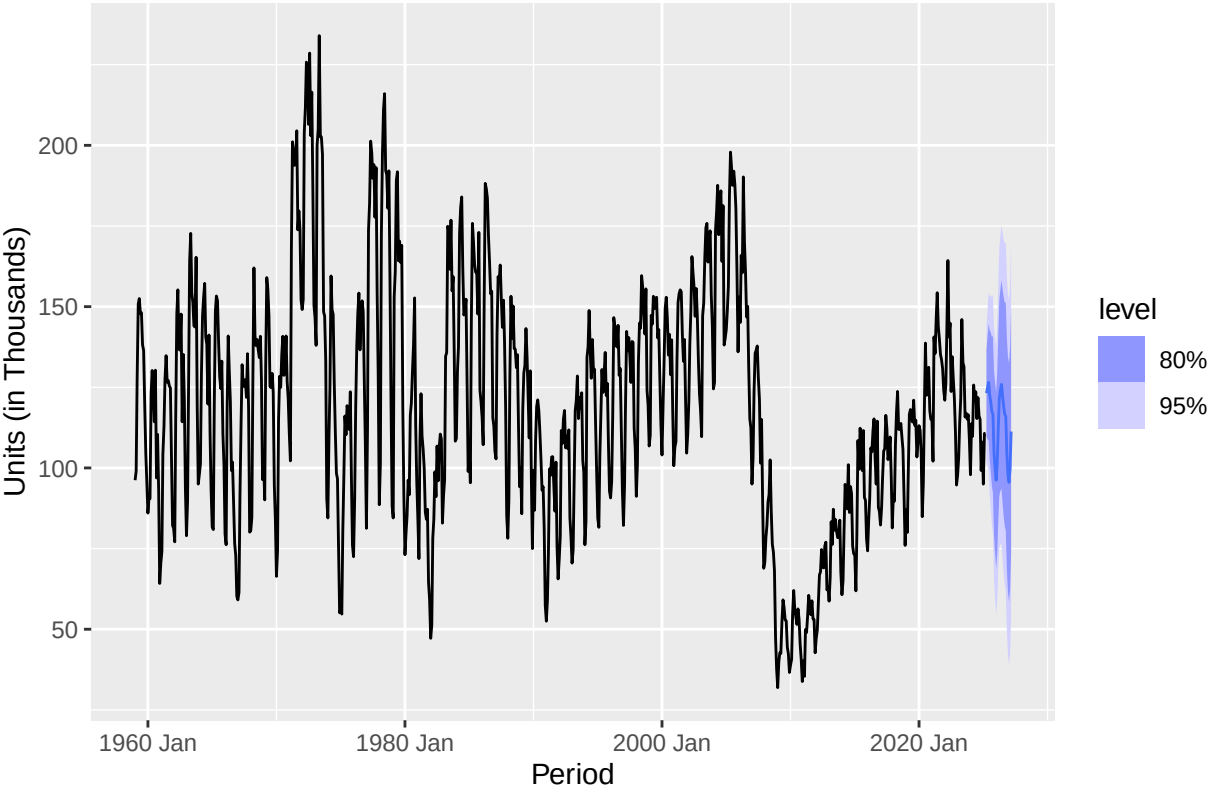
.model	.type	RMSE	MAE	MAPE	MASE	RMSSE
ETS(Value)	Training	10.327666	6.038569	7.150786	0.430289	9.4125984
ETS(Value)	Test	21.659416	7.468066	4.570816	0.935033	6.8653113
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	Training	10.354006	6.093471	7.485970	0.433228	7.4136510
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	Test	23.484014	9.683295	6.239130	1.053610	7.9382055
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	Training	10.106627	7.948910	7.355477	0.425490	6.4037678
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	Test	61.023783	22.919924	4.109772	2.832706	2.4379500
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + Case.Shiller + pdq())	Training	8.145118	6.344075	6.755125	0.466409	4.4556236
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + Case.Shiller + pdq())	Test	18.327022	4.135503	1.603605	0.756647	6.7321795

The best model was the regression with lagged predictors and Case-Shiller. All of the models performed better on this test set compared to the 80/20 split.

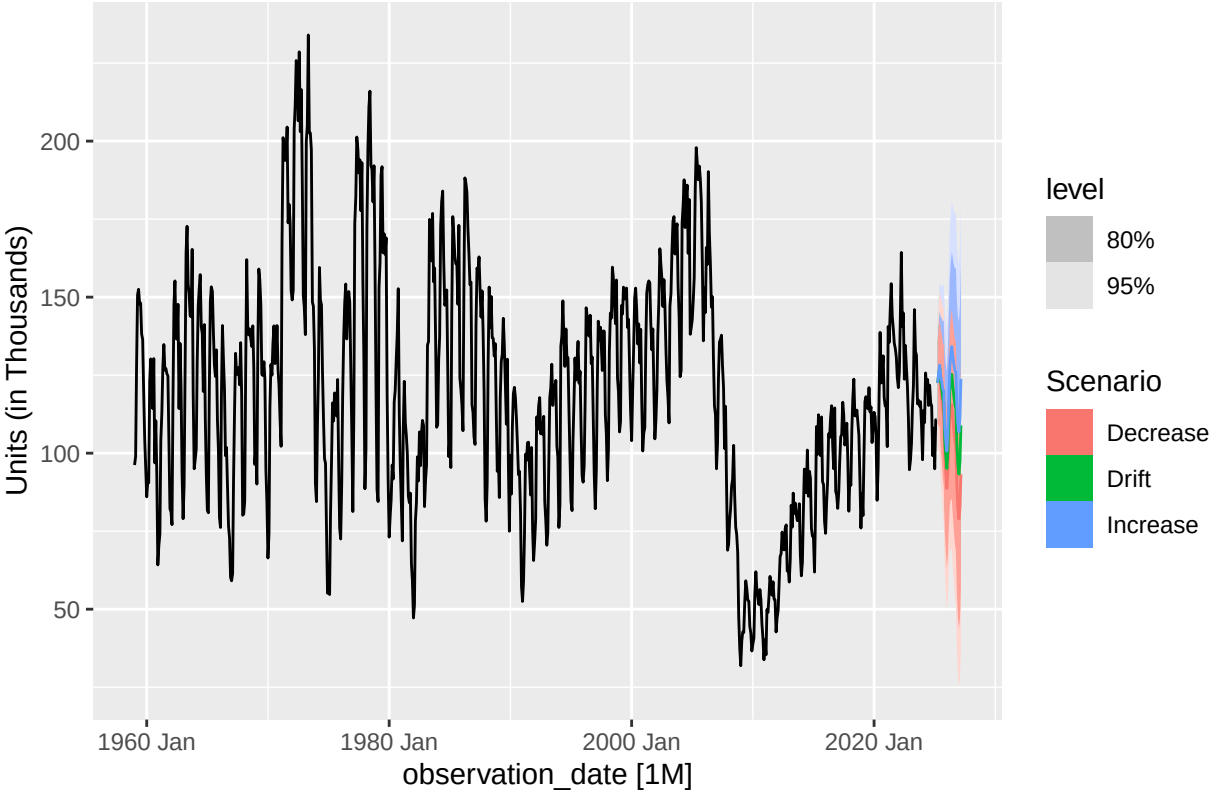
Privately-Owned Housing Units Started ETS 2 Yrs Forecast



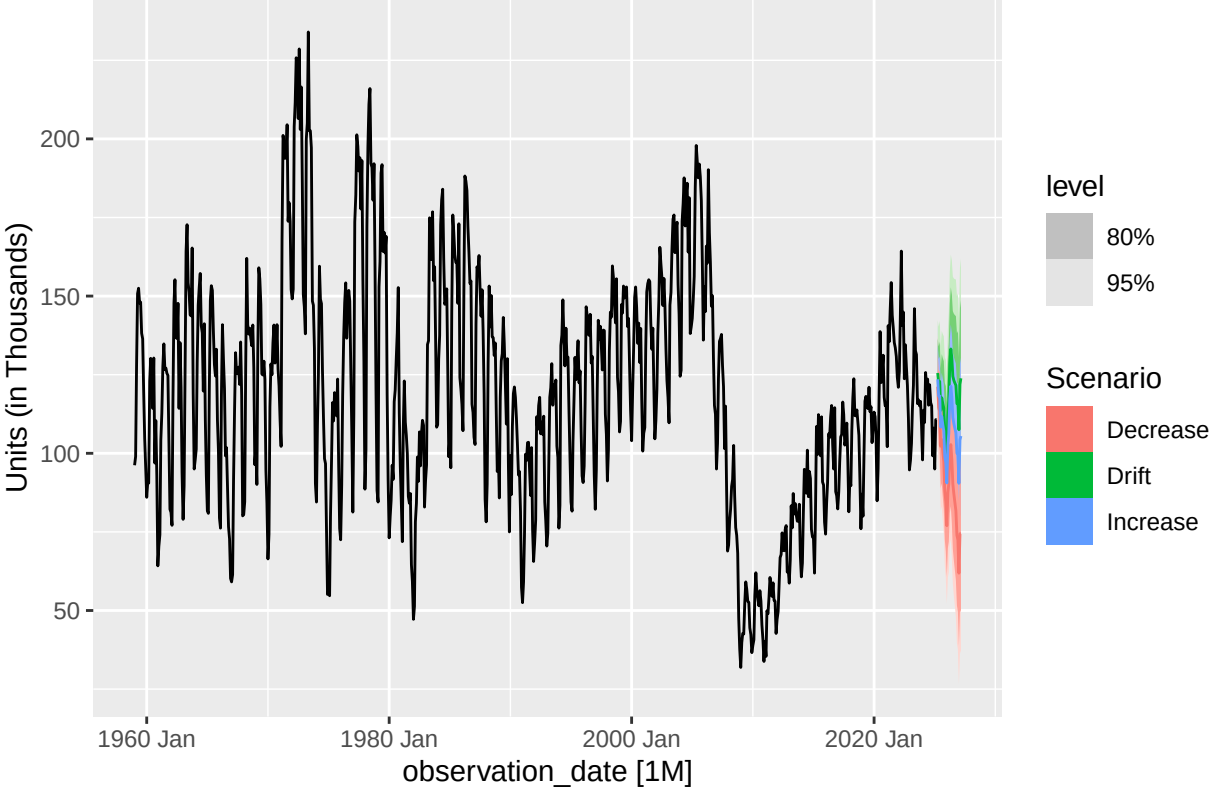
Two Year Forecasts
Privately-Owned Housing Units Started ARIMA 1 Yr Forecast



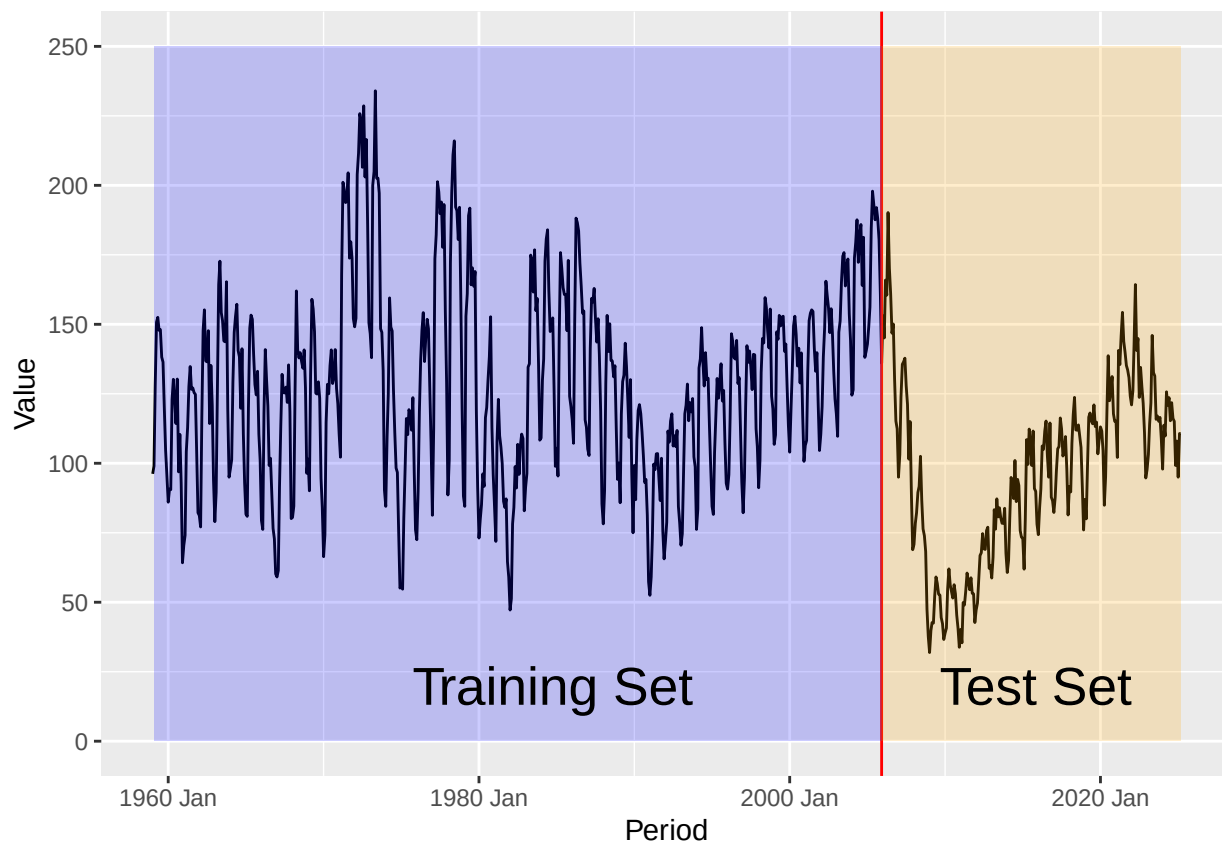
Privately-Owned Housing Units Started Regression1 1 Yr Forecast



Privately-Owned Housing Units Started Regression2 1 Yr Forecast



70/30 Training/Test Set



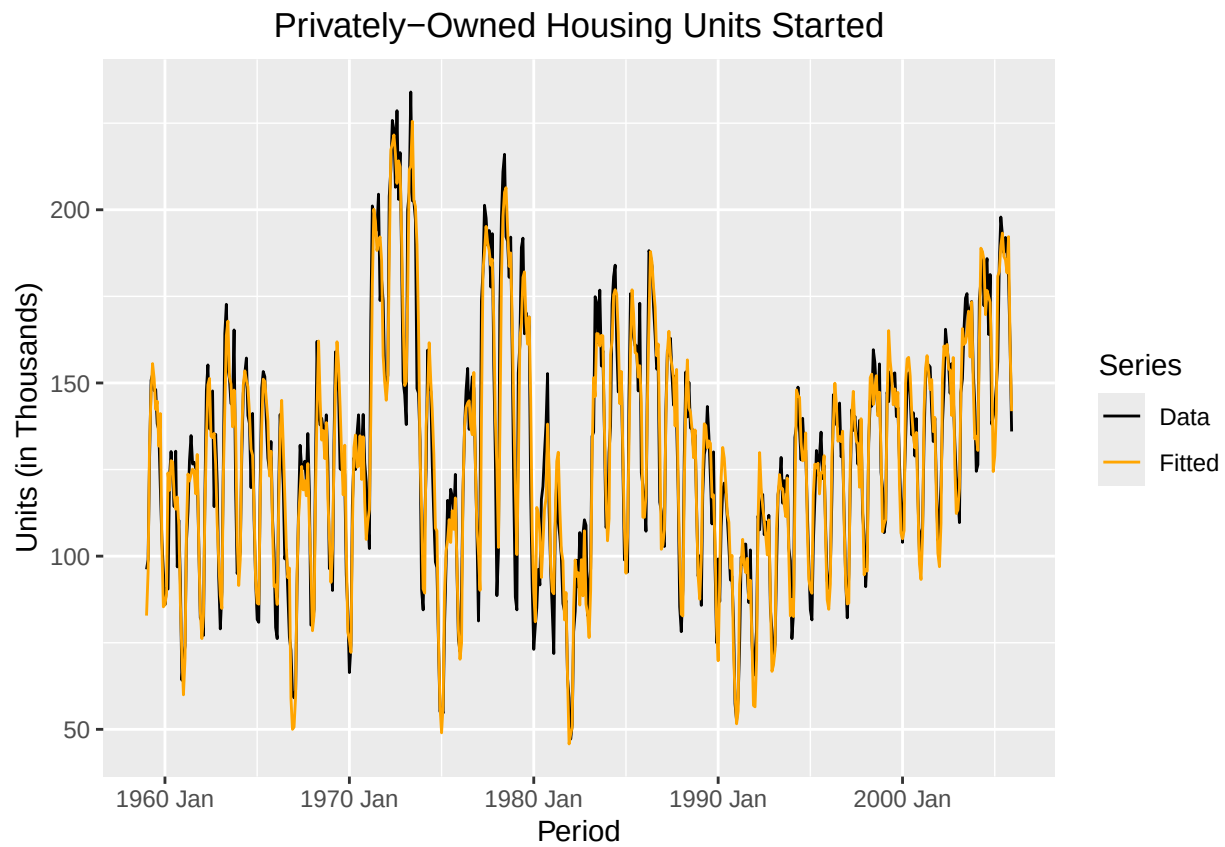
ETS

ETS Fit

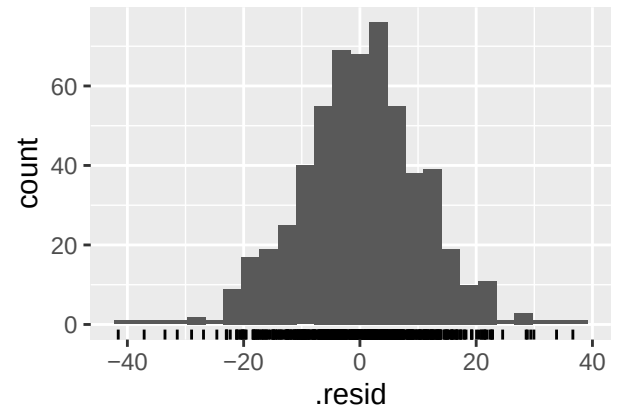
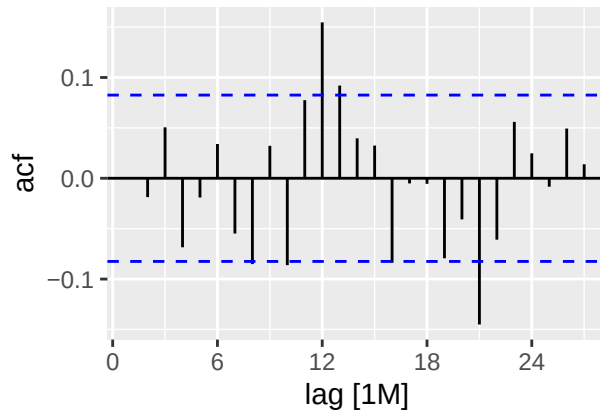
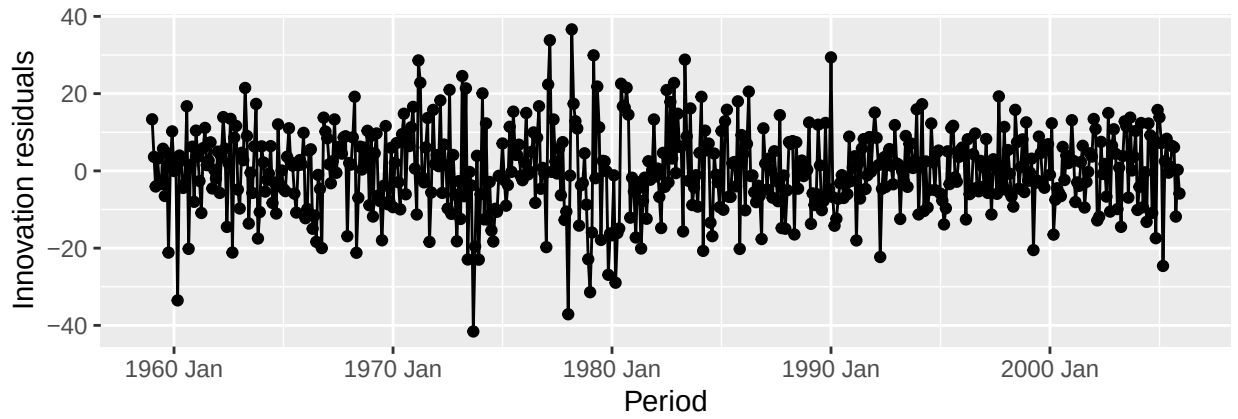
```
## Series: Value
## Model: ETS(A,N,A)
## Smoothing parameters:
##   alpha = 0.7078359
##   gamma = 0.0001000037
##
## Initial states:
##   l[0]    s[0]    s[-1]    s[-2]    s[-3]    s[-4]    s[-5]    s[-6]
## 119.761 -30.3461 -11.58074 11.84934 5.733311 14.05438 15.77951 22.20014
##   s[-7]    s[-8]    s[-9]    s[-10]    s[-11]
## 24.41113 18.69182 -0.0375639 -33.84125 -36.91398
##
## sigma^2: 117.3672
##
##      AIC      AICc      BIC
## 6276.427 6277.303 6341.453
```

ETS(Value)

<ETS(A,N,A)>



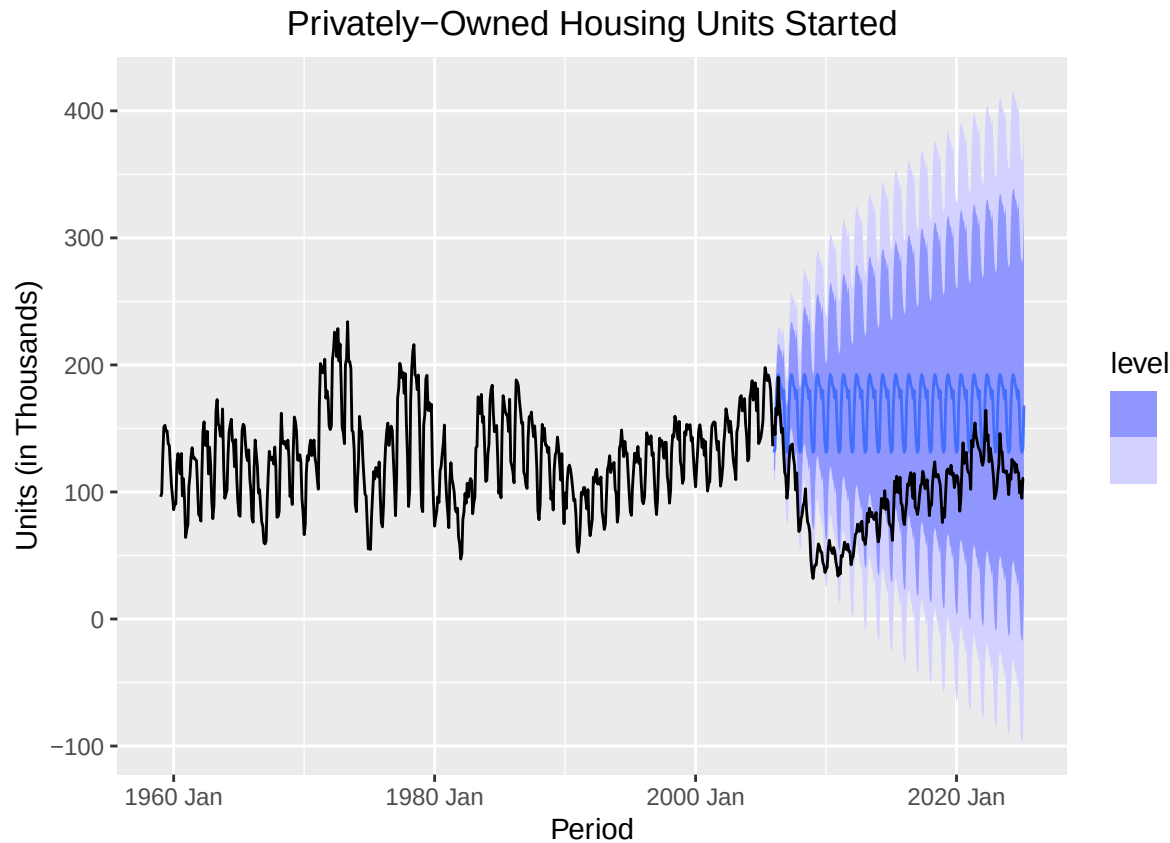
The ETS function selected a additive error, no trend component, and a additive seasonal component. The gamma term is similar to the ETS model for the 80/20 split. Alpha is around the same value as the last two splits.



Residuals

.model	lb_stat	lb_pvalue
ETS(Value)	65.24876	1.12e-05

The residuals are normally distributed around zero. There are clear spikes in the residuals but no overall trend. Prediction intervals may be robust. The autocorrelation of the residuals however does not resemble white noise. The Ljung-Box test clearly shows significance. The model does not include all relevant information and can be improved.



Test set Evaluation

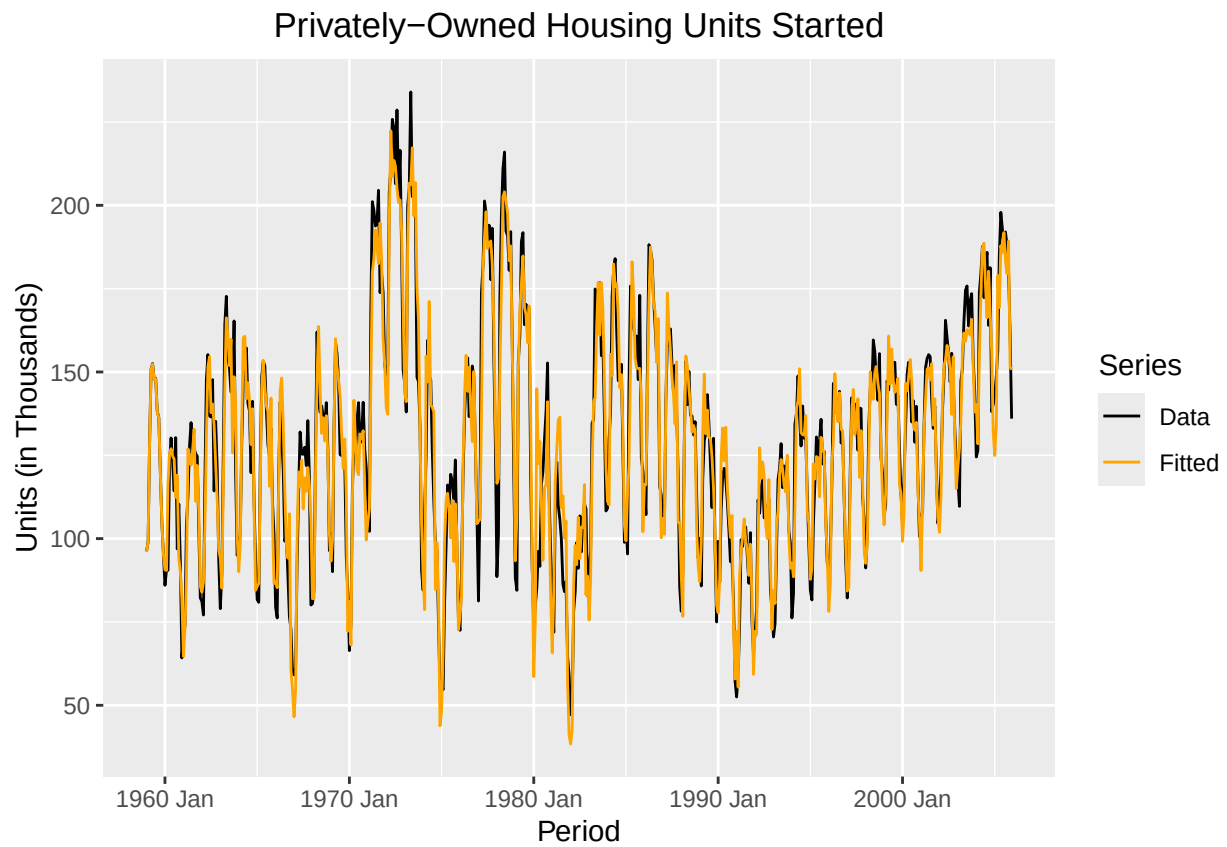
The ETS model completely misses the decline caused by the recession. It predicts a constant trend due to the lack of a trend component.

ARIMA

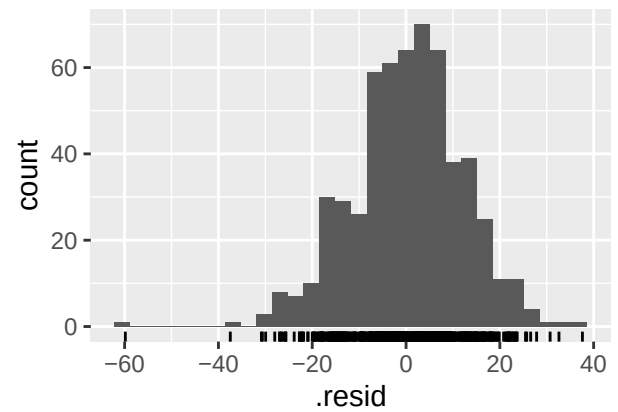
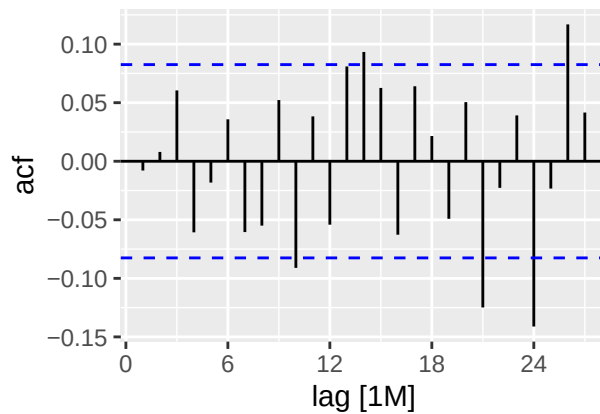
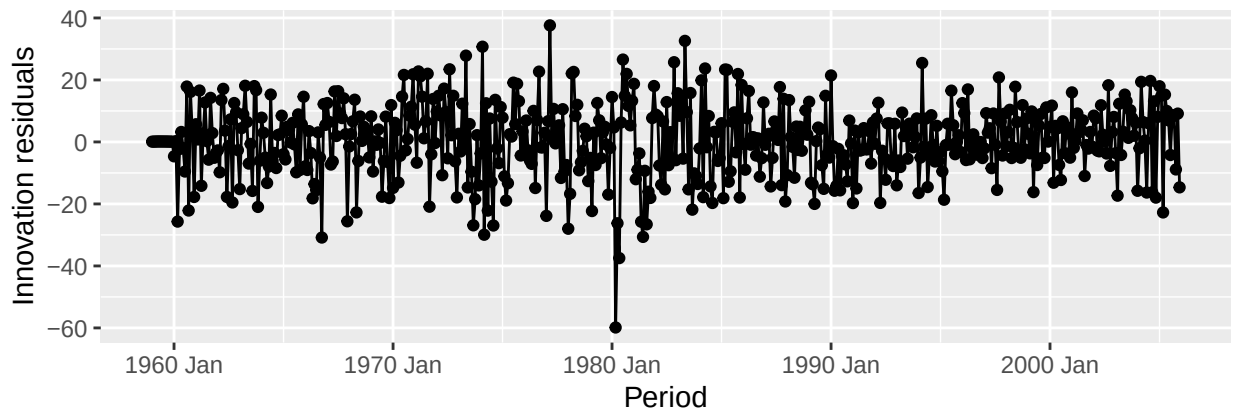
ARIMA Fit

```
## Series: Value
## Model: ARIMA(2,0,2)(2,1,0)[12]
##
## Coefficients:
##          ar1          ar2          ma1          ma2          sar1          sar2
##          1.9543      -0.9601      -1.3630      0.3824      -0.5116      -0.2127
## s.e.    0.0133      0.0130      0.0422      0.0419      0.0431      0.0432
##
## sigma^2 estimated as 143.1:  log likelihood=-2153.01
## AIC=4320.01   AICc=4320.22   BIC=4350.21
```

```
ARIMA(Value, stepwise = FALSE, approximation = FALSE)
<ARIMA(2,0,2)(2,1,0)[12]>
```



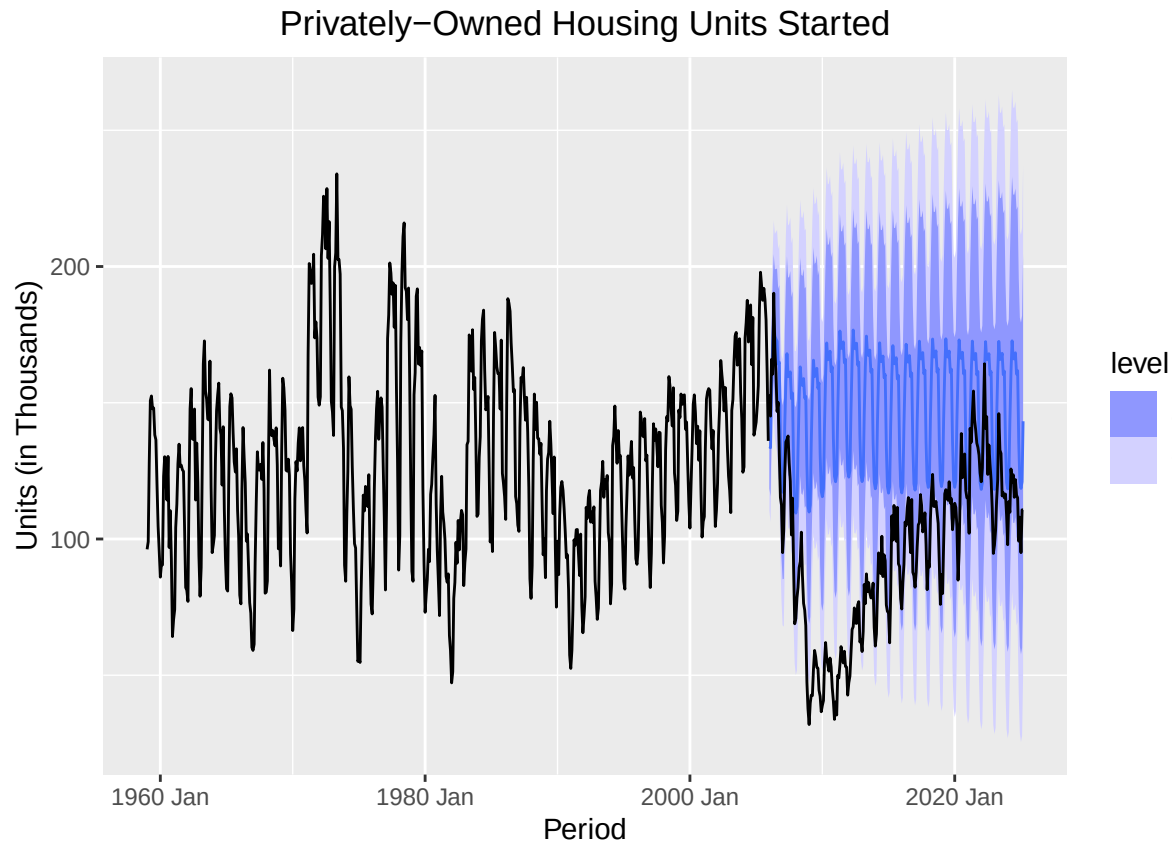
The ARIMA model has seasonal differencing. There is no moving average component in the seasonal part.



Residuals

.model	lb_stat	lb_pvalue
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	59.04506	1.78e-05

The residuals are normally distributed around zero. There are clear spikes in the residuals. There is a especially large residual around January 1980. However, there is no overall trend. Prediction intervals may be robust. The autocorrelation of the residuals however does not resemble white noise. The Ljung-Box test clearly shows significance. The model does not include all relevant information and can be improved.



Test Set Evaluation

ARIMA performs slightly better than ETS. However, it still fails to predict the decline during the recession.

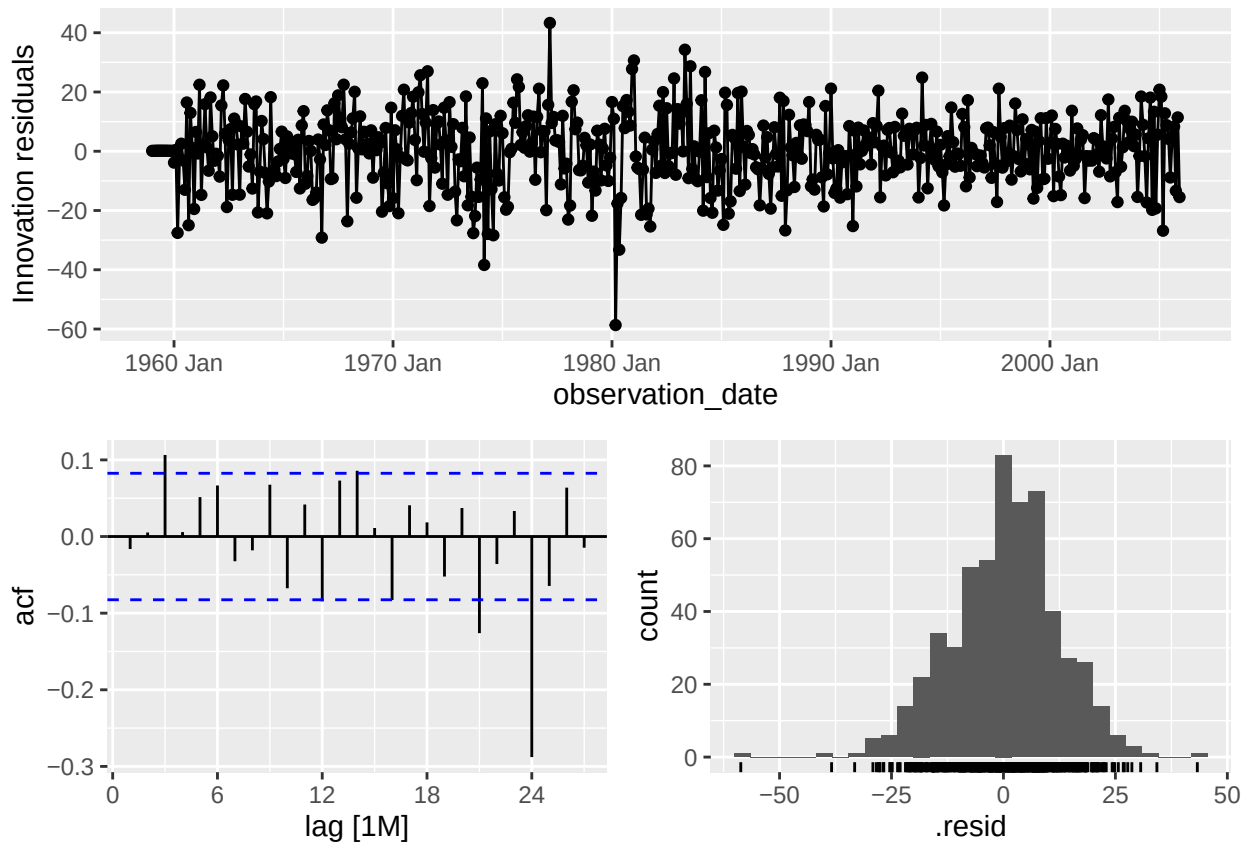
Regression with ARIMA Errors

Regression Fit

```
##
## =====
##               Model 1      Model 2      Model 3      Model 4
## -----
## ar1           0.94 ***      0.95 ***      0.41         0.69
##               (0.02)      (0.02)      (0.82)      (0.59)
## ma1          -0.33 ***      -0.36 ***      -1.04         -1.35 *
##               (0.04)      (0.04)      (0.81)      (0.56)
## sar1         -0.42 ***      -0.40 ***      -0.59 ***      -0.42 ***
##               (0.04)      (0.04)      (0.07)      (0.07)
## UNRATE        -3.46         -2.48
##               (2.18)         (3.99)
## FEDFUNDS     -1.92 **       -3.79 *
##               (0.69)         (1.77)
## PPI_Construction 0.21         -0.06
##               (0.56)         (0.48)
## UNRATE_Lag      -2.63         3.19
##               (2.09)         (4.02)
## FEDFUNDS_Lag   -3.29 ***      -2.39
##               (0.66)         (1.69)
## PPI_Construction_Lag -0.37         -0.68
```

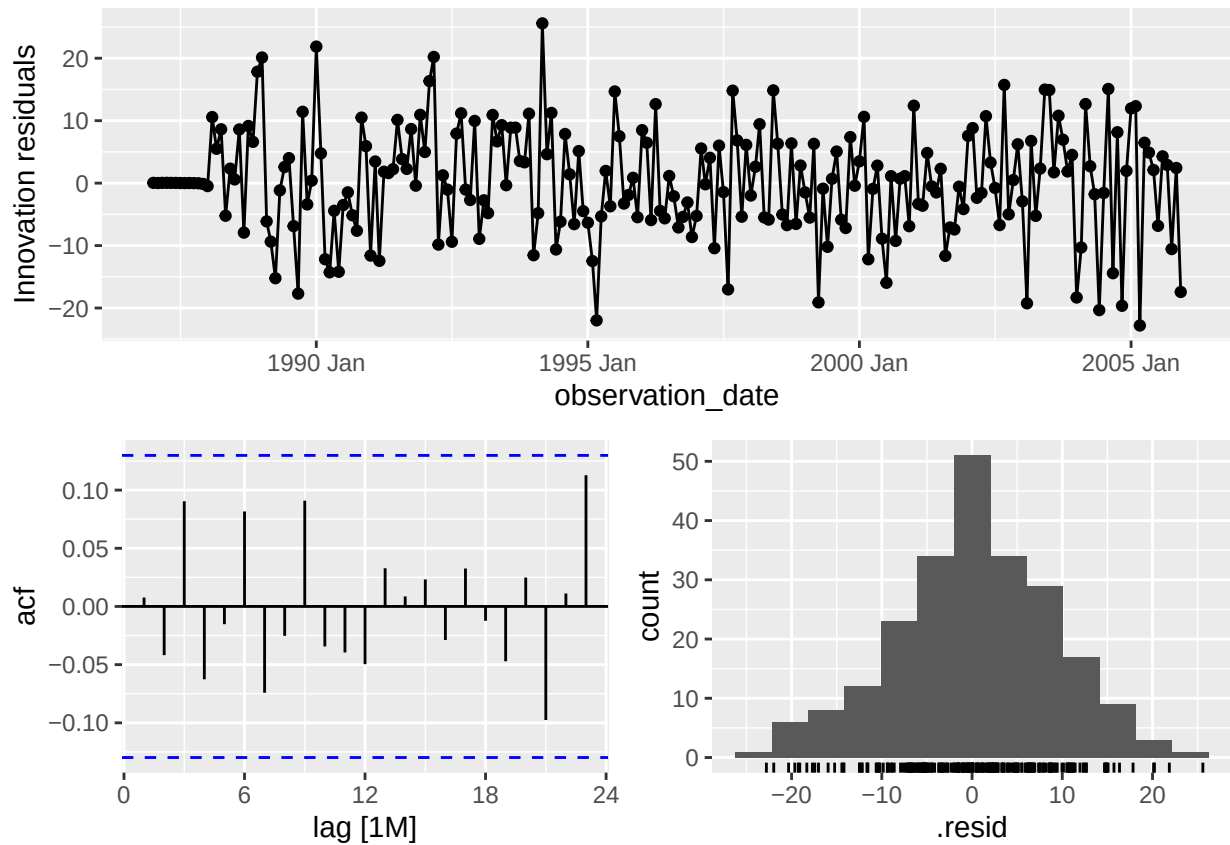
```
##                                (0.55)                                (0.54)
## ma2                                0.28                                0.49
##                                (0.49)                                (0.32)
## sar2                               -0.30 ***
##                                (0.07)
## Case.Shiller                        1.17                        2.33
##                                (1.01)                        (1.60)
## -----
## sigma2                        155.06                        150.22                        83.17                        88.80
## log_lik                       -2174.12                     -2165.19                     -778.45                     -784.70
## AIC                           4362.23                     4344.38                     1576.91                     1587.40
## AICc                           4362.44                     4344.59                     1577.99                     1588.28
## BIC                            4392.43                     4374.57                     1610.62                     1617.73
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

The best model were the regression with lag and the regression without lag for Case-Shiller. The latter model also performed the best.



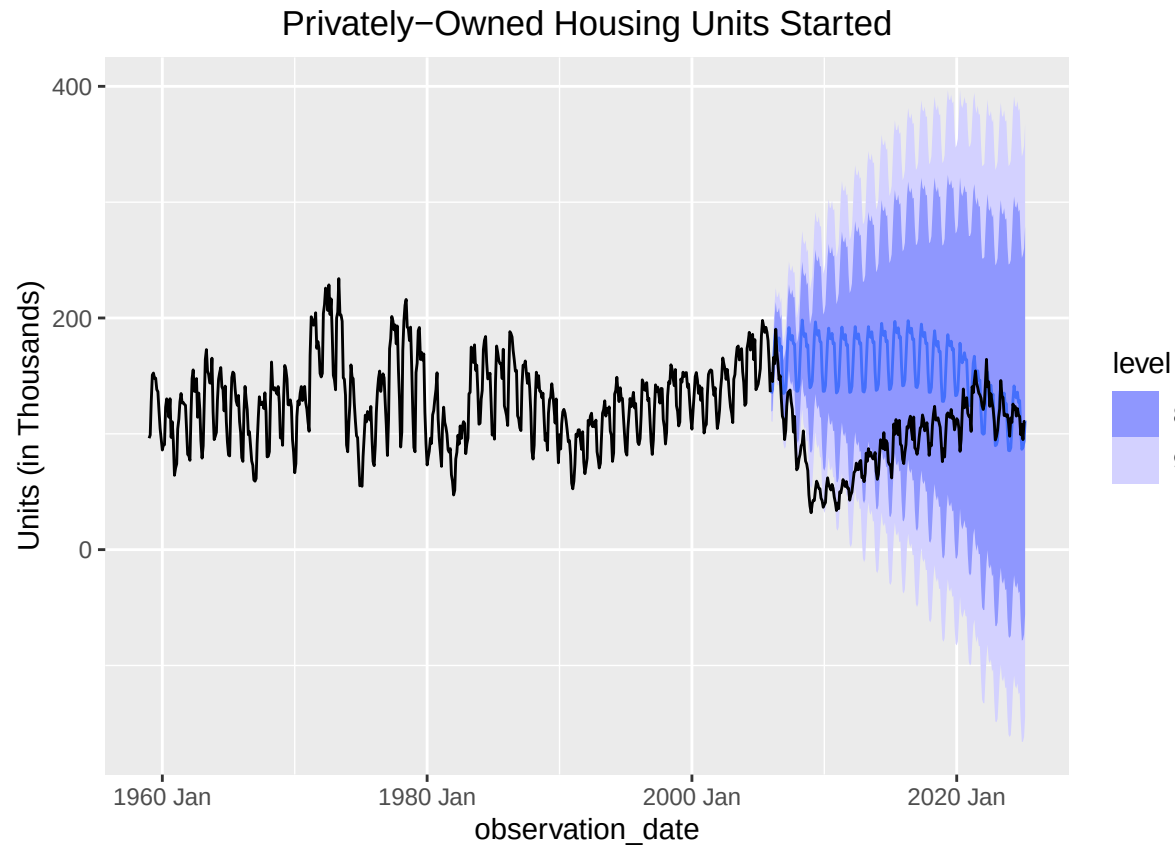
Residuals

.model	lb_stat	lb_pvalue
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq()))	96.59044	0

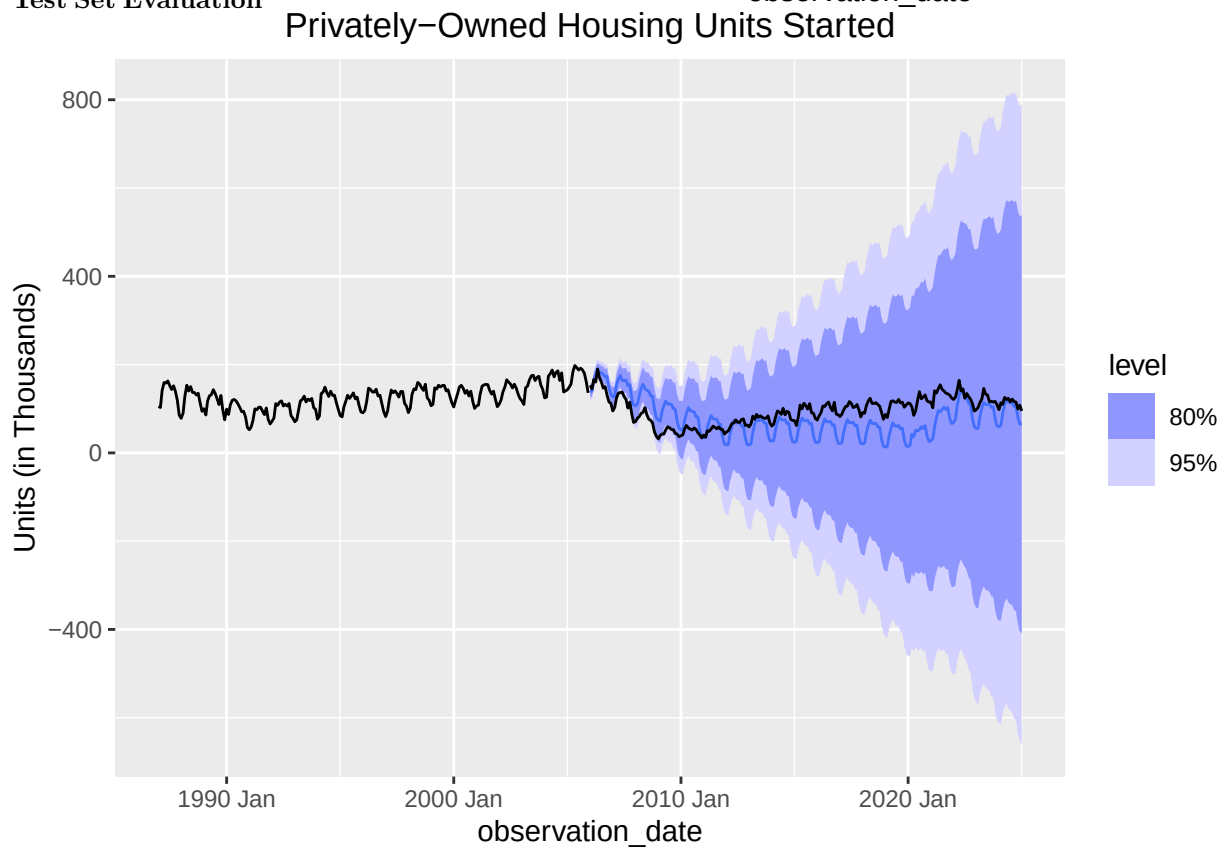


.model	lb_stat	lb_pvalue
ARIMA(Value ~ UNRATE + FEDFUNDS + PPI_Construction + Case.Shiller + pdq())	20.84644	0.4683558

Both regression have normally distributed residuals around zero. The case-shiller regression has many spikes. Unlike the regression without Case-Shiller, the second regression's ACF does resemble white noise. The Ljung-Box test shows no significance.



Test Set Evaluation



The first regression missed the decline in housing starts caused by the recession. The second regression

predicted the decline and recovery. However, it has much larger prediction intervals.

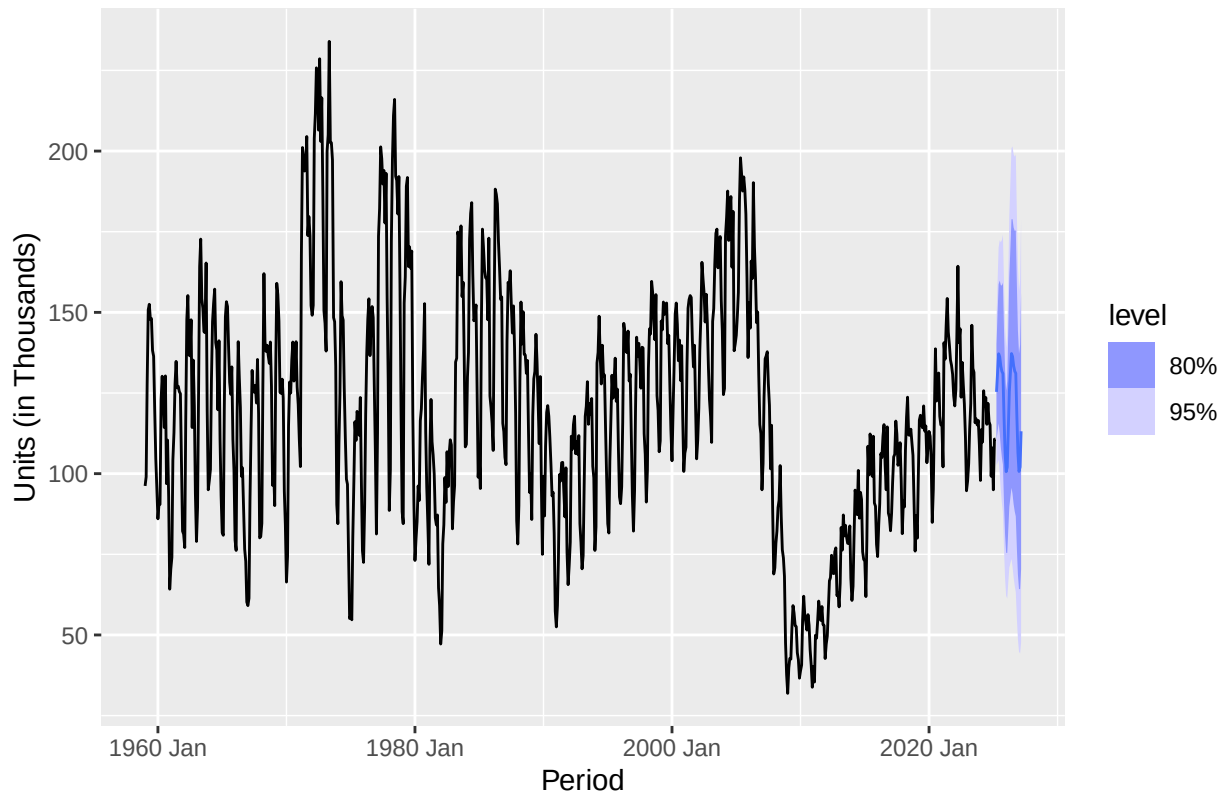
Model Accuracy

.model	.type	RMSE	MAE	MAPE	MASE	RMSSE
ETS(Value)	Training	10.698311	8.298690	6.845198	0.431755	0.4159406
ETS(Value)	Test	97.277540	89.836525	157.060013	3.673914	23.7820624
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	Training	11.770122	9.153963	7.745905	0.476252	0.4576116
ARIMA(Value, stepwise = FALSE, approximation = FALSE)	Test	79.131963	70.978732	126.995489	3.692802	13.0765788
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	Training	12.059152	9.360824	7.890485	0.487014	0.4688489
ARIMA(Value ~ UNRATE_Lag + FEDFUNDS_Lag + PPI_Construction_Lag + pdq())	Test	75.253020	63.891575	91.045066	3.324079	32.9257691
ARIMA(Value ~ UNRATE + FEDFUNDS + PPI_Construction + Case.Shiller + pdq())	Training	8.668576	6.769576	5.693838	0.583933	0.6062774
ARIMA(Value ~ UNRATE + FEDFUNDS + PPI_Construction + Case.Shiller + pdq())	Test	42.881470	37.312147	42.973297	1.941234	61.6671926

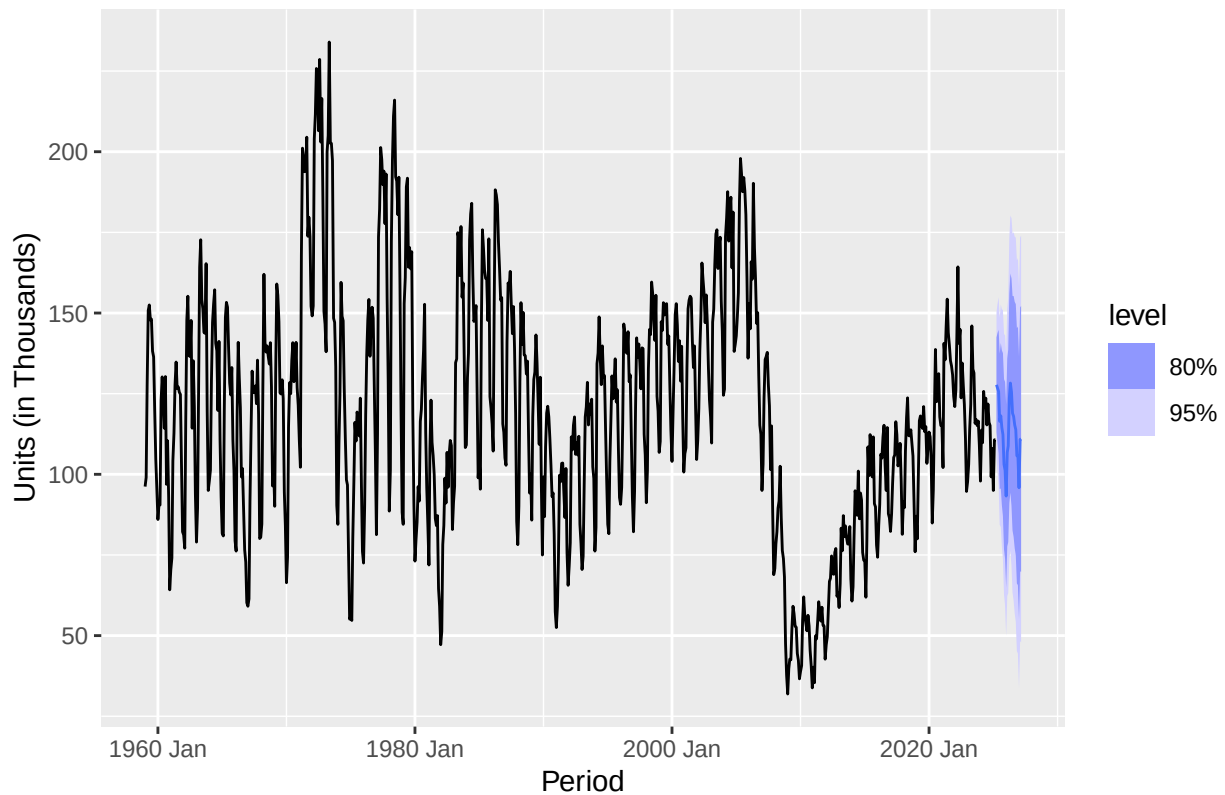
The best model is the regression with Case-Shiller and no lagged predictors. All models performed worse at predicting this test set compared to the other splits.

Two Year Forecasts

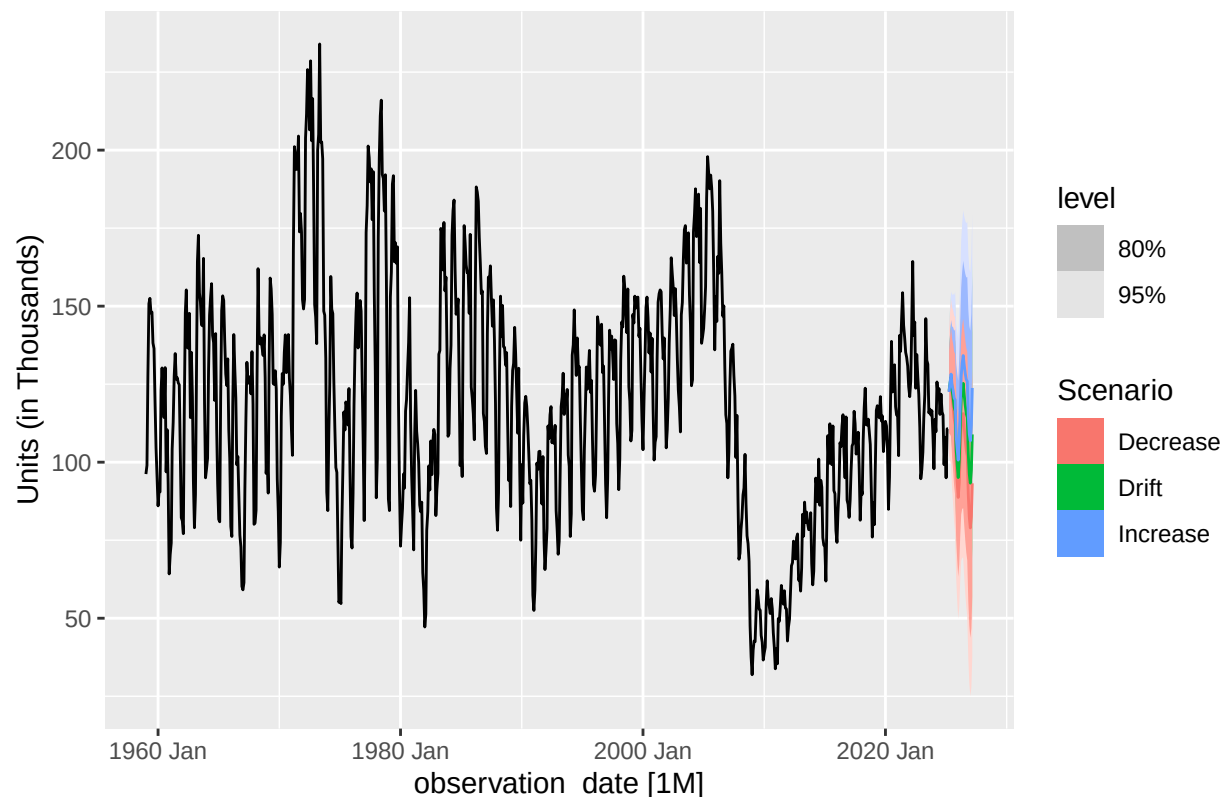
Privately-Owned Housing Units Started ETS 2 Yrs Forecast



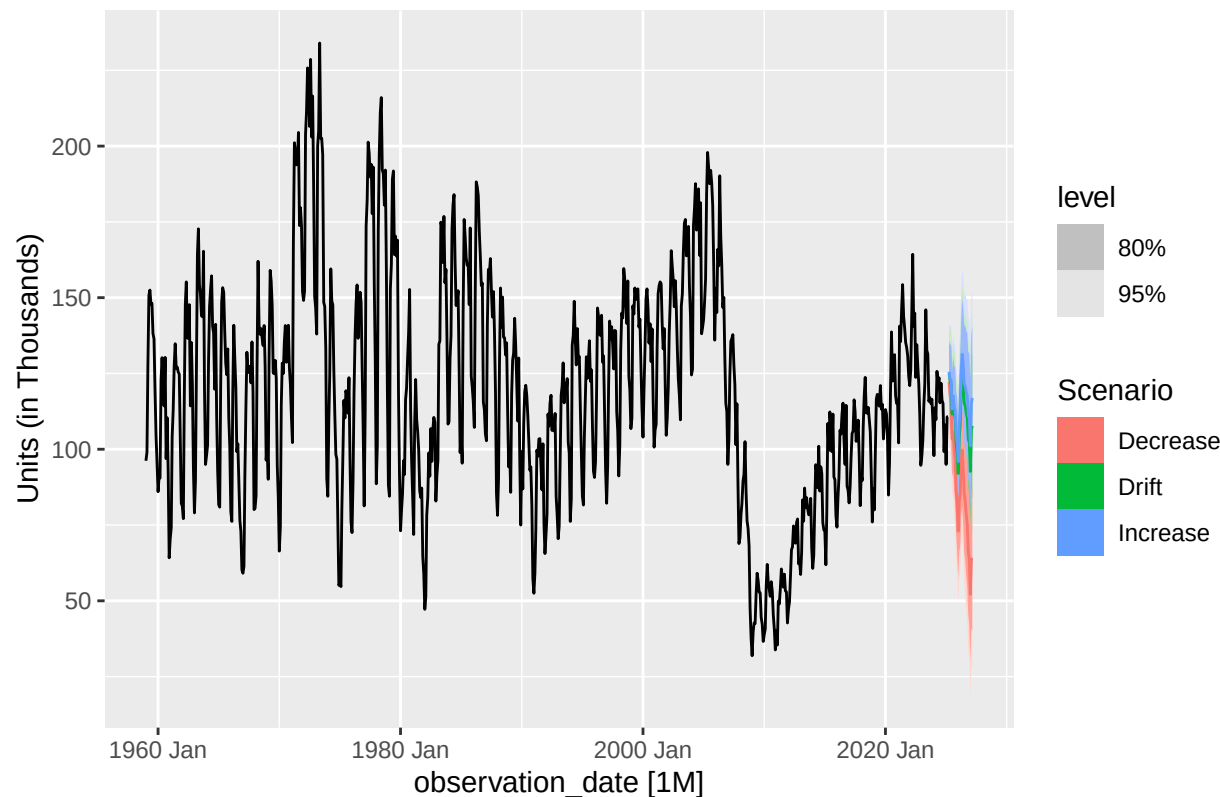
Privately-Owned Housing Units Started ARIMA 1 Yr Forecast



ely-Owned Housing Units Started Regression No Case-Shiller 2 Yrs Forecast



ely-Owned Housing Units Started Regression w/ Case-Shiller 2 Yrs Forecast



Conclusion

The best model in all of the training/test splits was the regression that included the Case-Shiller index. The Ljung-Box test for all the other models suggested the ACF of residuals were not white noise and were consequently were missing key information. The only models to not show significance were the regression with Case-Shiller. It is possible the missing information was home prices. This make intuitive sense, home builders are concerned with profit. While the regression with Case-Shiller performed well, it still struggled with predicting the decline in housing starts during the recession. All of the models struggled predicting extreme events. This is why models train on data which included the decline and recovery (90/10) had less error. In any case, the best models at forecasting housing starts were regression which included data on home prices. Housing prices thus is a key predictor for any future forecasts of housing construction.